

RTL Design Sherpa

STREAM Subsystem Hardware Architecture Specification

January 3, 2026

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2 Document Information

2.1 STREAM Hardware Architecture Specification

Document Number: STREAM-HAS-001 **Version:** 0.98 **Status:** Draft **Classification:** Open Source - MIT License

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2.2 Document Purpose

This Hardware Architecture Specification (HAS) provides a high-level architectural overview of the STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory) subsystem. It describes the system-level design, external interfaces, performance characteristics, and integration requirements without detailing internal implementation specifics.

Target Audience: - System architects evaluating STREAM for integration - Hardware engineers planning system-level integration - Software engineers developing drivers and firmware - Verification engineers planning system-level testing

Companion Documents: - STREAM Micro-Architecture Specification (MAS) - Detailed block-level implementation - STREAM Product Requirements Document (PRD) - Requirements and rationale

---

2.3 References

Reference Documents

ID	Document	Description
[REF-1]	STREAM MAS v0.90	Micro-Architecture Specification
[REF-2]	STREAM PRD	Product Requirements Document
[REF-3]	ARM AMBA AXI4	AXI4 Protocol Specification
[REF-4]	ARM AMBA APB	APB Protocol Specification

2.4 Terminology

Terminology and Definitions

Term	Definition
AXI	Advanced eXtensible Interface - ARM AMBA high-performance bus
APB	Advanced Peripheral Bus - ARM AMBA low-power configuration bus
Beat	Single data transfer on AXI bus (one clock cycle of valid data)
Burst	Sequence of consecutive beats forming a single AXI transaction
Channel	Independent DMA transfer context (STREAM is built with 8 or 16 channels)
Descriptor	256-bit data structure defining a single DMA transfer operation
DMA	Direct Memory Access - data transfer without CPU involvement
HAS	Hardware Architecture Specification
MAS	Micro-Architecture Specification
MonBus	Monitor Bus - internal debug/trace event streaming interface
Scatter-Gather	DMA mode using linked descriptors for non-contiguous transfers

2.5 Revision History

Document Revision History

Version	Date	Author	Description
0.90	2026-01-03	seang	Initial HAS release
0.91	2026-05-14	seang	Sync to RTL changes since 2026-04-17 (commit `be4e5a91`); regenerate PDFs/DOCX.

Version	Date	Author	Description
0.92	2026-06-05	seang	Sync to RTL state at 2026-06-05 (17 commits since `be4e5a91`). Documents (1) the 64→128-bit monbus packet migration + new 64-bit side-band timestamp wire — referenced via `docs/markdown/rtl-amba/includes/ocelus_package_spec.md`, with `cfg_ts_append_*` removed and m_axil records locked to 3 beats; (2) new APB channel-observation register set (`OBS_CTRL` / `OBS_FLAGS` / `OBS_DATA0` / `OBS_DATA1`) that exposes scheduler error stickies + timeout per channel; (3) BOTH-end descriptor-path monitoring with distinct `(UNIT_ID, AGENT_ID)` for fetch-side vs. consume-side, under the new per-port + global SV-parameter monitor methodology; (4) three RTL fixes — drain-ctrl stale-view race + post-flop wvalid gate (`a82627af`), registered `w_arb_request` for 8-channel timing closure (`4e8f9e02`), reset on SRAM avail outputs (`b619eee9`); (5) `stream_core_mon` duplicate cleanup (`7291c4ef`).

Version	Date	Author	Description
0.93	2026-07-02	seang	Monitor registers relocated to a separate `stream_mon_regs` regfile at 0x1000 (single APB slave); APB address decode widened to 13 bits / 8 KB so the monitor block is addressable (`ch04_interfaces/02_apb4_slave.md`). Companion MAS documents `SCHED_TIMEOUT_LIMIT` + the recoverable scheduler write-progress timeout and functional descriptor prefetch.
0.94	2026-07-07	seang	Documented the kick-burst fast-path interface (top-level `i_kick_burst_mask` / `i_kick_burst_addr` ports) that starts any subset of channels on a single clock cycle, bypassing the serial APB `CHn_CTRL` kick — added the interface signal table + semantics to `ch04_interfaces/02_apb4_slave.md`. Reference integration is the NexysA7 char harness (`CH_KICK_ADDR` shadow registers + `KICK_GO` bitmask CSR).
0.95	2026-07-09	seang	Documented read-ahead descriptor prefetch: the scheduler read side

Version	Date	Author	Description
			advances to the next chained legacy descriptor while the write side drains the current one, eliminating the per-descriptor boundary bubble (cross-descriptor streaming). Added a section to `ch05_performance/01_throughput.md` with on-silicon A/B (Nexys A7, single bitstream: 95.3% datapath util prefetch-on vs 76.1% off at 64-beat descriptors; row-major EXT 99.7% util bubble-free; transpose single-beat latency-bound). Runtime `SCHED_CONFIG.RD_PREFETCH_EN`, default on.
0.96	2026-10-17	seang	8- or 16-channel builds (`stream_top_ch8` / `stream_top_ch16`) and the 16-channel register map; one shared descriptor engine by default (`SHARED_ENGINES`), with measured area (8 channels 25.8k -> 20.7k LUTs; 16 channels 36.8k) in `ch05_performance/03_resources.md`. Corrected three pages that did not match the RTL: the channel FSM (six states, read and write concurrent in

Version	Date	Author	Description
			CH_XFER_DATA), SRAM buffering (`SRAM_DEPTH` is per channel, with no dynamic reallocation), and interrupts (one `stream_irq`, no per-channel vector). Diagrams ch02_system_context, ch03_whole_unit, ch03_channel_architecture and 05_channel_fsm redrawn.
0.97	2026-10-17	seang	Measured Genesys 2 board builds with `SHARED_ENGINES` 0 and 1: post-route utilization and timing in `ch05_performance/03_resources.md` (STREAM 30,781 -> 29,247 LUTs, 23,079 -> 20,654 flip-flops; BRAM and DSP unchanged), and the 40-configuration board sweep of both in `ch05_performance/01_throughput.md` (40 of 40 on both, line rate on both, the shared build one cycle sooner per run).
0.98	2026-10-20	seang	Block diagrams redrawn to the repo element key (v3): class fills, double-bordered registered blocks, green return paths and a generated key under each figure. No technical content

Version	Date	Author	Description
			changed.

Last Updated: 2026-10-17

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3 Purpose and Scope

3.1 Document Purpose

This Hardware Architecture Specification (HAS) defines the high-level architecture of the STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory) subsystem. It serves as the primary reference for:

- **System Integration** - Understanding STREAM's external interfaces and system-level requirements
- **Performance Planning** - Evaluating throughput, latency, and resource characteristics
- **Driver Development** - Programming model and software interface requirements
- **Verification Planning** - System-level test scenarios and coverage requirements

This document complements the STREAM Micro-Architecture Specification (MAS), which provides detailed block-level implementation specifics.

3.2 Scope

3.2.1 In Scope

This specification covers:

1. **System Architecture**
   - Block-level functional partitioning
   - Data flow through the subsystem
   - Multi-channel operation model
2. **External Interfaces**
   - APB configuration slave interface
   - AXI4 master interfaces (descriptor fetch, data read, data write)
   - Monitor bus (MonBus) output interface
3. **Performance Characteristics**
   - Throughput targets and limiting factors
   - Latency breakdown for typical operations
   - Resource estimates for FPGA/ASIC targets

4. **Integration Requirements**
   - Clock and reset requirements
   - System-level constraints
   - Verification strategy overview

3.2.2 Out of Scope

This specification does not cover:

- **Detailed Micro-Architecture** - See STREAM MAS for block-level implementation
- **RTL Implementation Details** - Register-level timing, pipeline stages
- **Testbench Architecture** - See verification documentation
- **Software Driver Implementation** - See software programming guide

---

3.3 Design Philosophy

3.3.1 Tutorial-Focused Simplicity

STREAM is intentionally designed as a beginner-friendly DMA engine tutorial. Key simplifications include:

Aspect	Simplification	Rationale
Address Alignment	Aligned addresses only	Eliminates complex fixup logic
Transfer Length	Specified in beats	Simplifies datapath math
Descriptor Chaining	Linear chains only	No circular buffer management
Flow Control	Simple transaction limits	No credit management complexity

These simplifications make STREAM an ideal learning platform while maintaining production-quality RTL practices.

3.3.2 Relationship to RAPIDS

STREAM is derived from the RAPIDS (Rapid AXI Programmable In-band Descriptor System) architecture but removes:

- Network interfaces (source/sink paths)
- Address alignment fixup logic
- Credit-based flow control
- Control read/write engines

This creates a pure memory-to-memory DMA engine suitable for understanding core DMA concepts.

3.4 Intended Audience

Audience	Primary Use
System Architects	Evaluate STREAM for system integration
Hardware Engineers	Plan physical integration and timing closure
Software Engineers	Develop drivers and firmware
Verification Engineers	Plan system-level test coverage

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4 Document Conventions

4.1 Notation Conventions

4.1.1 Signal Naming

Prefix/Suffix	Meaning	Example
`i_`	Input signal	`i_clk`, `i_data`
`o_`	Output signal	`o_valid`, `o_ready`
`r_`	Registered signal	`r_state`, `r_count`
`w_`	Combinational wire	`w_next_state`
`_n`	Active-low signal	`aresetn`, `rst_n`

4.1.2 Interface Prefixes

Prefix	Protocol	Direction
`s_apb_`	APB slave	Inbound
`m_axi_desc_`	AXI4 descriptor master	Outbound
`m_axi_rd_`	AXI4 read master	Outbound
`m_axi_wr_`	AXI4 write master	Outbound
`monbus_`	Monitor bus	Outbound

4.2 Diagram Conventions

4.2.1 Block Diagrams

- **Rectangles** - Functional blocks
- **Arrows** - Data flow direction
- **Dashed Lines** - Configuration/control paths
- **Double Lines** - High-bandwidth data paths

4.2.2 Timing Diagrams

- **High** - Logic 1
- **Low** - Logic 0
- **X** - Don't care / undefined
- **Hatched** - Valid data window

4.3 Numeric Conventions

4.3.1 Units

Unit	Meaning
Beat	Single data transfer (one clock cycle of valid data)
Burst	Sequence of consecutive beats
Transaction	Complete AXI transaction (address + all data beats)

4.3.2 Address Notation

- All addresses are specified in bytes unless otherwise noted
- Address alignment requirements are specified relative to data width
- Example: 512-bit data width requires 64-byte (512/8) alignment

4.3.3 Size Notation

Notation	Value
KB	1,024 bytes
MB	1,048,576 bytes
Kbit	1,024 bits

4.4 Requirement Notation

Requirements and constraints are identified using the following format:

Prefix	Category
[REQ-x]	Functional requirement
[PERF-x]	Performance requirement
[INTEG-x]	Integration requirement
[CONST-x]	Design constraint

4.5 Reference Format

Cross-references to other documents use the format:

- **[MAS-x.y]** - Reference to MAS section x.y
- **[REF-n]** - Reference to document in reference table

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5 Definitions and Acronyms

5.1 Acronyms

Acronym	Definition
APB	Advanced Peripheral Bus
AXI	Advanced eXtensible Interface
AMBA	Advanced Microcontroller Bus Architecture
BFM	Bus Functional Model
CDC	Clock Domain Crossing
DMA	Direct Memory Access
FIFO	First-In First-Out buffer
FSM	Finite State Machine
HAS	Hardware Architecture Specification
MAS	Micro-Architecture Specification
MonBus	Monitor Bus (internal debug/trace interface)
PRD	Product Requirements Document
RTL	Register Transfer Level
SG	Scatter-Gather
SRAM	Static Random Access Memory
STREAM	Scatter-gather Transfer Rapid Engine for AXI Memory

5.2 Definitions

5.2.1 Architectural Terms

Term	Definition
Beat	A single data transfer on an AXI bus, corresponding to one clock cycle where valid data is transferred.

Term	Definition
	For a 512-bit data bus, one beat transfers 64 bytes.
Burst	A sequence of consecutive beats forming a single AXI transaction. STREAM supports burst lengths up to 256 beats (AXI4 maximum).
Channel	An independent DMA transfer context. STREAM is built with 8 or 16 channels, each capable of processing its own descriptor chain.
Descriptor	A 256-bit data structure in memory that defines a single DMA transfer operation, including source address, destination address, length, and chaining information.
Descriptor Chain	A linked list of descriptors where each descriptor points to the next, enabling complex scatter-gather transfer patterns.
Scatter-Gather	A DMA technique where data is transferred to/from non-contiguous memory locations using a list of descriptors.

5.2.2 Interface Terms

Term	Definition
APB Slave	STREAM's configuration interface, receiving programming commands from the system processor.
AXI Master	STREAM's memory interfaces for descriptor fetches and data transfers. Three masters: descriptor fetch, data read, data write.
MonBus	Monitor Bus - a 64-bit streaming interface for debug and performance monitoring packets.
Backpressure	Flow control mechanism where a receiver signals inability to accept data, causing the sender to pause.

5.2.3 Transfer Terms

Term	Definition
Aligned Address	An address that is a multiple of the data width in

Term	Definition
	bytes. For 512-bit (64-byte) data width, aligned addresses are multiples of 64.
Kick-off	Initiating a DMA transfer by writing the first descriptor address to a channel's control register.
Completion	The successful end of a descriptor chain, optionally generating an interrupt.
Chaining	Following the `next_descriptor_ptr` to fetch and process subsequent descriptors.

5.3 Units and Measurements

Term	Definition
Clock Cycle	One period of the primary system clock (aclk).
Latency	Time from initiating an operation to its completion, typically measured in clock cycles.
Throughput	Data transfer rate, typically measured in bytes per second or beats per cycle.
Utilization	Percentage of available bandwidth actually used for data transfer.

5.4 Document-Specific Terms

Term	Definition
External Interface	A signal or bus that crosses the STREAM subsystem boundary.
Internal Interface	A signal or bus between blocks within STREAM, not visible externally.
Configuration Register	An APB-accessible register for controlling STREAM behavior.
Status Register	An APB-readable register reporting STREAM operational state.

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6 Use Cases

6.1 Primary Use Cases

6.1.1 Memory-to-Memory Data Movement

Description: Transfer data between two memory regions without CPU involvement.

Operation: 1. Software prepares descriptor(s) in memory with source/destination addresses 2. Software writes first descriptor address to channel control register 3. STREAM autonomously fetches descriptor and executes transfer 4. Optional interrupt notifies software of completion

Key Benefits: - CPU offloading for bulk data transfers - Deterministic transfer timing - Support for large transfer sizes via descriptor chaining

---

6.1.2 Scatter-Gather Operations

Description: Collect data from multiple non-contiguous source regions into a contiguous destination, or distribute contiguous source data to multiple destinations.

Gather Example:

```
Descriptor 0: src=0x1000, dst=0x8000, len=64 beats
Descriptor 1: src=0x3000, dst=0x8100, len=32 beats  (chained)
Descriptor 2: src=0x5000, dst=0x8180, len=16 beats  (chained, last)
```

Scatter Example:

```
Descriptor 0: src=0x8000, dst=0x1000, len=64 beats
Descriptor 1: src=0x8100, dst=0x3000, len=32 beats  (chained)
Descriptor 2: src=0x8180, dst=0x5000, len=16 beats  (chained, last)
```

6.1.3 Multi-Channel Concurrent Transfers

Description: Execute multiple independent transfer operations simultaneously.

Operation: - Up to 8 or 16 channels (by top) operate concurrently - Each channel processes its own descriptor chain - Shared AXI masters picked round-robin; each channel has its own SRAM FIFO

Use Cases: - Parallel DMA for multi-stream applications - Priority-based bandwidth allocation - Independent channel error isolation

---

6.2 Secondary Use Cases

6.2.1 Buffer Management

Description: Move data between system memory and local buffers for processing pipelines.

Pattern:

```
System Memory → STREAM → Processing Buffer → STREAM → System Memory
```

6.2.2 Memory Initialization

Description: Fill memory regions with pattern data using chained descriptors.

Note: Requires source buffer containing pattern data.

6.3 Target Applications

Application	Channel Usage	Typical Transfer Size
Video Frame Copy	1-2 channels	1-8 MB per frame
Audio Buffer Management	1 channel	4-64 KB per buffer
Data Collection	4-8 channels	Variable
Memory Test Patterns	1-4 channels	System-dependent

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7 Key Features

7.1 Feature Summary

Feature	Specification
Channels	8 or 16 independent channels (`stream_top_ch8` / `stream_top_ch16`)
Descriptor Size	256 bits (32 bytes)
Data Width	Parameterizable (default 512 bits)
Address Width	64 bits
Maximum Burst	256 beats (AXI4)
Chaining	Unlimited descriptor chain depth
Arbitration	Priority-based with round-robin

7.2 Descriptor-Based Operation

7.2.1 Single-Write Kick-off

A single APB write to a channel's control register initiates an entire transfer sequence:

1. Write descriptor address to `CHn_CTRL` register
2. STREAM automatically fetches and processes descriptor
3. Chained descriptors processed without software intervention
4. Completion indicated via status register and optional interrupt

7.2.2 Autonomous Chaining

STREAM follows `next_descriptor_ptr` automatically:

- **Non-zero pointer**: Fetch next descriptor, continue processing
- **Zero pointer**: Chain complete, transition to idle
- **Last flag**: Explicit chain termination regardless of pointer

7.3 Multi-Channel Architecture

7.3.1 Channel Independence

Each channel maintains:

- Independent FSM state
- Separate descriptor chain pointer
- Individual error status
- Private completion interrupt

7.3.2 Resource Sharing

All channels share:

- Descriptor fetch AXI master
- Data read AXI master
- Data write AXI master
- Internal SRAM buffer
- MonBus reporter

7.3.3 Priority-Based Arbitration

- 8-bit priority field in each descriptor
- Higher priority descriptors serviced first
- Round-robin within same priority level
- Anti-starvation timeout mechanism

7.4 Data Path Features

7.4.1 Parameterizable Data Width

Compile-time configurable data width:

Parameter	Description	Typical Values
`DATA_WIDTH`	AXI data bus width	128, 256, 512 bits
`SRAM_DEPTH`	Internal buffer depth	1024, 2048, 4096 entries

7.4.2 Aligned Address Requirement

[CONST-1] All source and destination addresses must be aligned to data width.

Data Width	Alignment Requirement
128 bits	16-byte aligned
256 bits	32-byte aligned
512 bits	64-byte aligned

This simplification eliminates complex alignment fixup logic for tutorial clarity.

7.5 Monitoring and Debug

7.5.1 MonBus Integration

64-bit monitor packets for:

- Transfer initiation events
- Descriptor fetch events
- Transfer completion events
- Error condition reporting
- Performance metrics

7.5.2 Status Visibility

APB-readable status registers provide:

- Channel state (idle, active, error)
- Descriptor count per channel
- Error codes and flags
- Performance counters (optional)

7.6 Error Handling

7.6.1 Error Detection

Error Type	Detection	Response
AXI read error	RRESP != OKAY	Stop chain, set error flag

Error Type	Detection	Response
AXI write error	BRESP != OKAY	Stop chain, set error flag
Invalid descriptor	Validation failure	Stop chain, set error flag
Address out of range	Limit check	Stop chain, set error flag

7.6.2 Error Isolation

Errors in one channel do not affect other channels. Each channel has independent error status and recovery.

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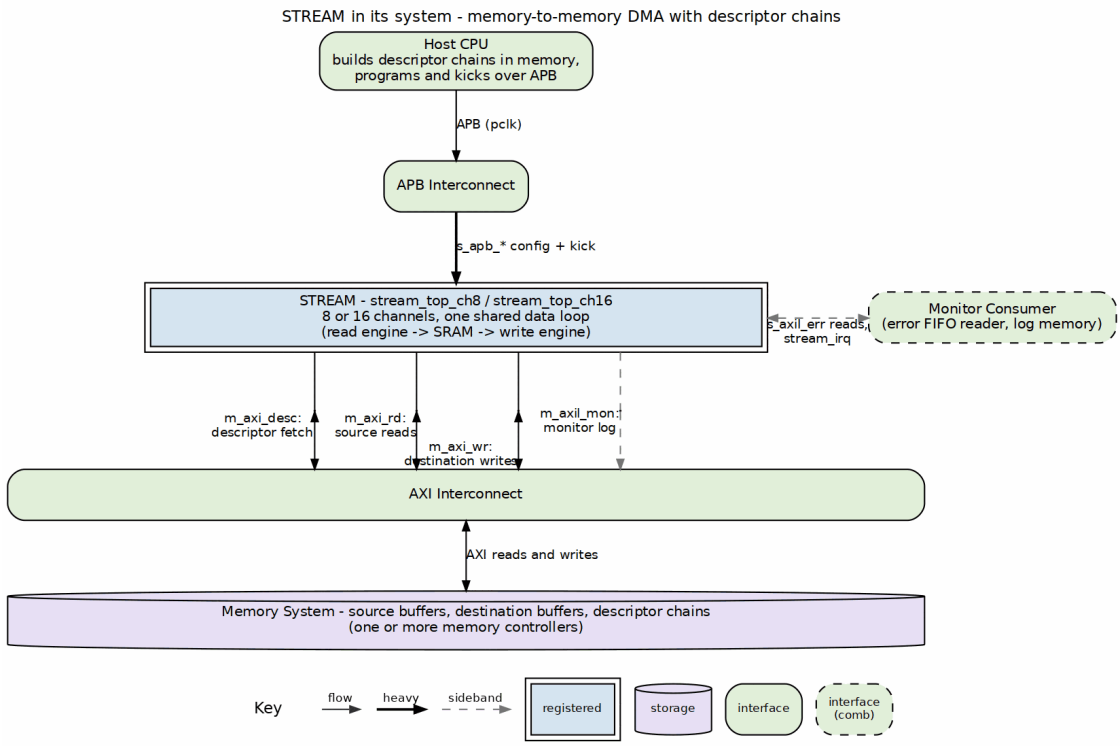
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8 System Context

8.1 System Integration Overview

STREAM integrates into a typical SoC as a peripheral subsystem with the following connections:

8.1.1 Figure 2.3.1: System Context



System Context

Source: ch02_system_context.dot

8.2 External Connections

8.2.1 Configuration Path

Connection	Protocol	Description
APB Slave	APB3/APB4	Configuration and control interface

Functions: - Channel kick-off (write descriptor address) - Status monitoring - Interrupt control - Error handling

8.2.2 Data Paths

Connection	Protocol	Width	Description
Descriptor Master	AXI4	256-bit	Fetch descriptors from memory
Read Data Master	AXI4	Param	Read source data from memory
Write Data Master	AXI4	Param	Write destination data to memory

8.2.3 Monitoring Path

Connection	Protocol	Width	Description
MonBus Master	MonBus	64-bit	Debug/trace packet output

8.3 Clock and Reset

8.3.1 Clock Domain

STREAM operates in a single clock domain:

Signal	Description
`aclk`	Primary system clock, all interfaces synchronous
`aresetn`	Active-low asynchronous reset

[INTEG-1] All external interfaces must be synchronous to `aclk`.

8.3.2 Reset Requirements

- Asynchronous assertion, synchronous de-assertion
- Minimum reset pulse width: 2 clock cycles
- All state machines return to idle on reset
- SRAM contents undefined after reset

8.4 Memory Map Requirements

8.4.1 Descriptor Memory

Descriptors must be placed in memory accessible via the descriptor fetch AXI master:

- **Alignment:** 32-byte aligned (256-bit descriptor size)
- **Accessibility:** Read access required
- **Coherency:** Software must ensure descriptors are coherent before kick-off

8.4.2 Data Memory

Source and destination regions must be accessible via the appropriate AXI master:

- **Source:** Read access via read data master
- **Destination:** Write access via write data master
- **Alignment:** Per data width (see Key Features)

8.5 Interrupt Integration

8.5.1 Interrupt Signals

Interrupt Signals

Signal	Description
`stream_irq`	The single interrupt output: the error/interrupt FIFO of `monbus_axil4_axil4_group` is not empty. Tied to 0 when `USE_AXI_MONITORS = 0`

There is no per-channel interrupt vector, at 8 or at 16 channels.

8.5.2 Interrupt Conditions

- A MonBus packet the group classifies as an error or interrupt (for example a descriptor completion with the descriptor's `interrupt` flag, or a channel error) is pushed into the error FIFO, which raises `stream_irq`
- Software reads the FIFO through `s_axil_err` to find the channel and cause, and the interrupt drops when the FIFO is empty
- Per-channel completion is also visible by polling `AXI_RD_COMPLETE`, `AXI_WR_COMPLETE` and `CHANNEL_IDLE`

8.6 Power Considerations

8.6.1 Clock Gating

STREAM supports clock gating when all channels are idle:

- Monitor `all_channels_idle` status bit
- External clock gate can disable `aclk` when idle
- Re-enable clock before writing kick-off register

8.6.2 Power States

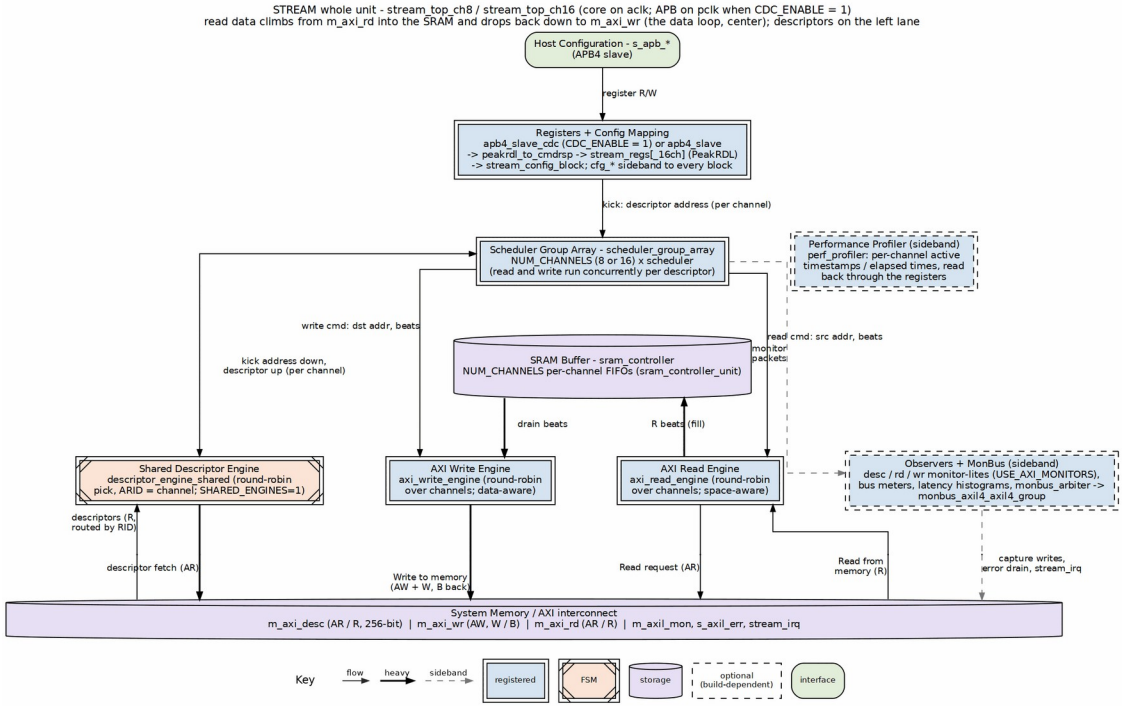
State	Channels	Clock	Power
Active	>= 1 busy	Running	Full
Idle	All idle	Running	Reduced (no switching)
Gated	All idle	Stopped	Minimal (leakage only)

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9 Block Diagram

9.1 Top-Level Architecture

9.1.1 Figure 3.1.1: STREAM Whole Unit



STREAM Whole Unit

Source: ch03_whole_unit.dot

9.2 Functional Block Summary

Block	Function	Key Interfaces
APB Config Slave + Registers	`apb4_slave_cdc` (or `apb4_slave`), `stream_regs` (`stream_regs_16ch` in the 16-channel top), `stream_config_block`	APB slave, kick-off signals
Descriptor Engine	Fetch and parse descriptors. One	AXI read, per-channel descriptor output

Block	Function	Key Interfaces
	shared engine serves every channel by default (SHARED_ENGINES = 1)	
Scheduler	Coordinate transfer phases	Control signals to all blocks
Descriptor AR Arbiter	SHARED_ENGINES = 0 only: round-robin over the per-channel descriptor engines (no priority). The shared engine makes the same pick internally	Grant signals
AXI Read Engine	Read source data to SRAM	AXI read master
SRAM Buffer	Temporary data storage	Read/write ports
AXI Write Engine	Write SRAM data to destination	AXI write master
Monitors + MonBus group	Monitor-lites, monbus_arbiter, monbus_axil4_axil4_group (USE_AXI_MONITORS = 1)	m_axil_mon, s_axil_err, stream_irq
Performance Profiler	Per-channel activity timestamps or elapsed times	Register read-back

9.3 Block Interactions

9.3.1 Control Flow

APB Write -> APB Config -> Descriptor Engine -> Scheduler -> Data Path, with AXI Desc Master fetching descriptors.

9.3.2 Data Flow

Memory -> AXI Read Engine -> SRAM Buffer -> AXI Write Engine -> Memory

9.3.3 Monitoring Flow

Scheduler groups and monitor-lites -> `monbus_arbiter` -> `monbus_axil4_axil4_group` -> error FIFO (`s_axil_err`, `stream_irq`) and capture writes (`m_axil_mon`)

9.4 Parameterization

Parameter	Description	Default	Range
`NUM_CHANNELS`	Number of DMA channels	8 (`stream_top_ch8`), 16 (`stream_top_ch16`)	1-16 (8-channel register map up to 8)
`SHARED_ENGINES`	One shared descriptor engine (1) or one per channel (0)	1	0, 1
`DATA_WIDTH`	AXI data bus width	512	128, 256, 512
`ADDR_WIDTH`	Address width	64	32, 64
`SRAM_DEPTH`	Buffer depth in entries, per channel	4096	256-8192
`DESC_FIFO_DEPTH`	Descriptor FIFO depth	4	2-8

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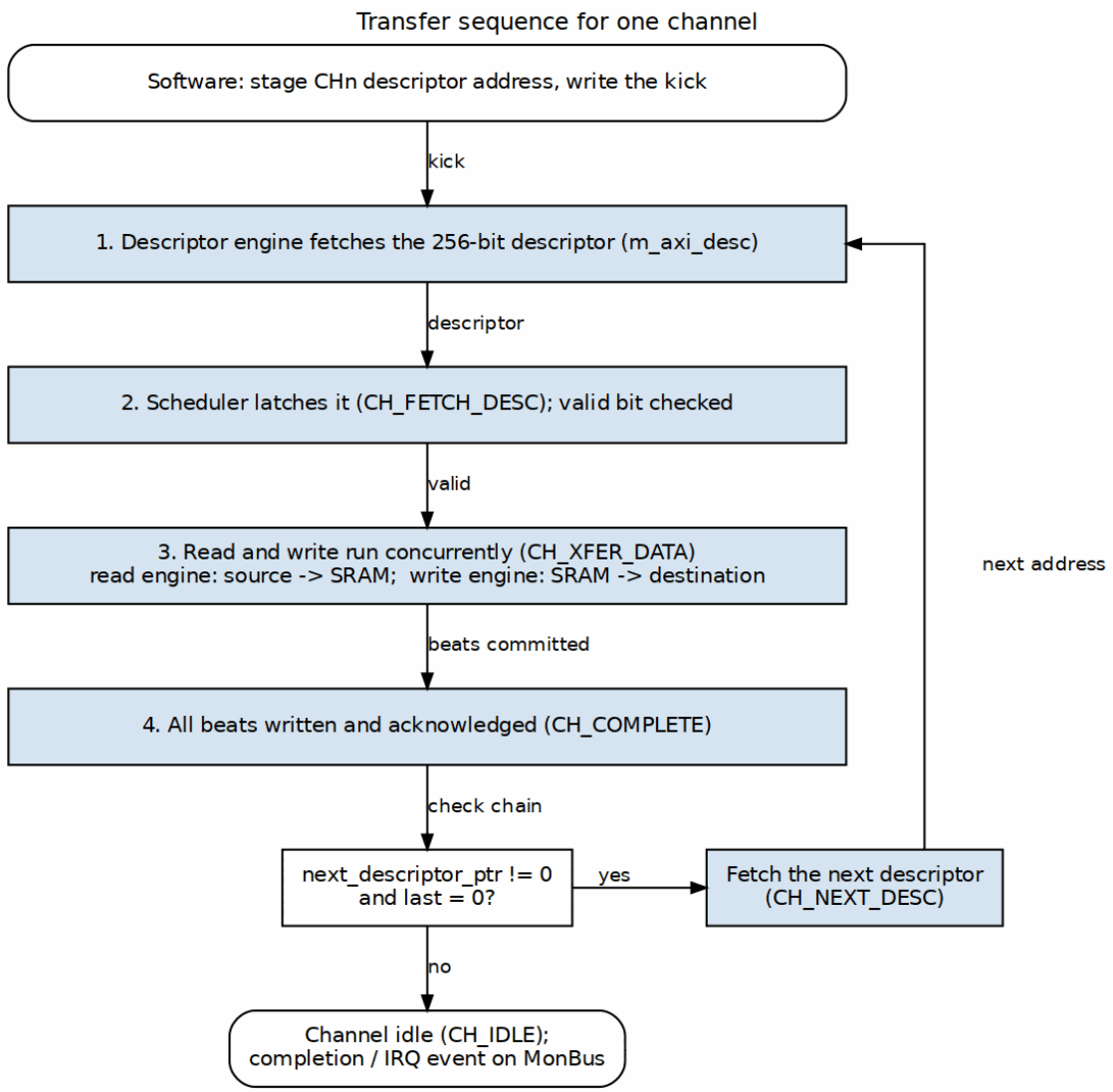
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10 Data Flow

10.1 Transfer Sequence Overview

A complete STREAM transfer follows this sequence:

10.1.1 Figure 3.2.1: Transfer Sequence



Transfer Sequence

Source: ch03_data_flow.dot

10.2 Phase Details

10.2.1 Phase 1: Kick-off

Software writes the first descriptor address to the channel control register:

- APB write triggers channel state transition from IDLE to ACTIVE
- Descriptor address stored in channel's descriptor pointer register
- Kick-off signal sent to Descriptor Engine

10.2.2 Phase 2: Descriptor Fetch

Descriptor Engine issues AXI read transaction:

- Address: From kick-off or `next_descriptor_ptr`
- Size: 256 bits (single beat or 2-beat depending on bus width)
- Response: Descriptor data captured on valid read data

10.2.3 Phase 3: Descriptor Parse

Descriptor fields extracted and validated:

Field	Validation
`valid`	Must be 1
`src_addr`	Alignment check, range check
`dst_addr`	Alignment check, range check
`length`	Non-zero, within limits

10.2.4 Phase 4: Arbitration

Channel requests access to shared data path:

- Multiple channels may be active simultaneously
- The read and write engines grant channels round-robin (no priority)
- Granted channel proceeds to read phase

10.2.5 Phase 5: Read Phase

AXI Read Engine fetches source data:

- Issues AXI read bursts to source address
- Data written to SRAM buffer
- Continues until `length` beats transferred

- Read/write to SRAM can overlap (ping-pong or streaming)

10.2.6 Phase 6: Write Phase

AXI Write Engine sends data to destination:

- Reads data from SRAM buffer
- Issues AXI write bursts to destination address
- Awaits write responses
- Continues until all data written

10.2.7 Phase 7: Chain Check

Scheduler determines next action:

- If `last == 1` or `next_descriptor_ptr == 0`: Chain complete
- Otherwise: Fetch next descriptor (return to Phase 2)

10.3 Concurrent Operation

10.3.1 Multi-Channel Concurrency

Multiple channels can be at different phases simultaneously. For example: - Ch0: Kick -> Fetch -> Read -> Write -> Complete - Ch1: (offset) Kick -> Fetch -> Arb Wait -> Read -> Write - Ch2: (offset) Kick -> Fetch -> Arb Wait -> Read

10.3.2 Pipeline Overlap

Within a single channel, read and write phases can overlap. As read beats 0-63 complete, write beats 0-63 can begin while read continues with beats 64-127.

The SRAM buffer enables this overlap by storing in-flight data.

10.4 Data Path Bandwidth

10.4.1 Theoretical Maximum

With 512-bit data width at 200 MHz:

- Per-direction: 512 bits x 200 MHz = 102.4 Gbps = 12.8 GB/s
- Bidirectional: 25.6 GB/s (simultaneous read + write)

10.4.2 Practical Considerations

Actual throughput affected by:

- AXI arbitration latency
- Memory access latency
- SRAM buffer depth (limits outstanding data)
- Channel arbitration overhead

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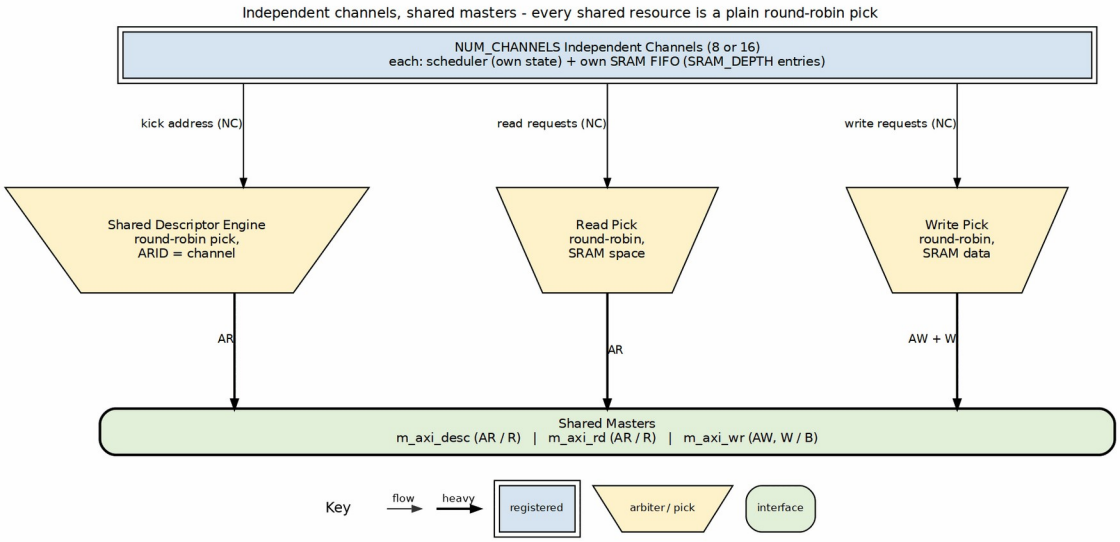
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11 Multi-Channel Architecture

11.1 Channel Organization

STREAM supports 8 or 16 independent DMA channels (`stream_top_ch8` / `stream_top_ch16`), each processing its own descriptor chain:

11.1.1 Figure 3.3.1: Channel Architecture



Channel Architecture

Source: ch03_channel_architecture.dot

11.2 Per-Channel Resources

Each channel maintains independent state:

Resource	Description	Size
FSM State	Current operation phase	4 bits
Descriptor Pointer	Current/next descriptor address	64 bits
Transfer Counter	Beats remaining in current transfer	32 bits
Error Status	Error flags and codes	8 bits

Resource	Description	Size
Priority	Current descriptor priority	8 bits

11.3 Shared Resources

All channels share the following resources:

Resource	Arbitration	Capacity
Descriptor AXI Master	Round-robin pick in the shared descriptor engine	One fetch in flight per channel; fetches from different channels overlap (ARID = channel)
Data Read AXI Master	Round-robin over channels with SRAM space	Configurable outstanding
Data Write AXI Master	Round-robin over channels with SRAM data	Configurable outstanding
SRAM Buffer	None: one FIFO per channel	SRAM_DEPTH entries per channel
MonBus	Round-robin (`monbus_arbiter`)	One packet per grant

11.4 Arbitration Scheme

11.4.1 Round-Robin Arbitration

Every shared master is behind a plain round-robin pick (`arbiter_round_robin` in acknowledge mode): the grant moves to the next requesting channel after the last winner. There is no priority logic. The descriptor's 8-bit priority field (`[207:200]`) is carried but not used for arbitration, and there is no anti-starvation timer or preemption; round-robin itself bounds every channel's wait to one turn of the other channels. The read and write engines mask a channel out of their pick until its SRAM FIFO has room (read) or data (write) for a burst.

11.4.2 Arbitration Points

Arbitration occurs at:

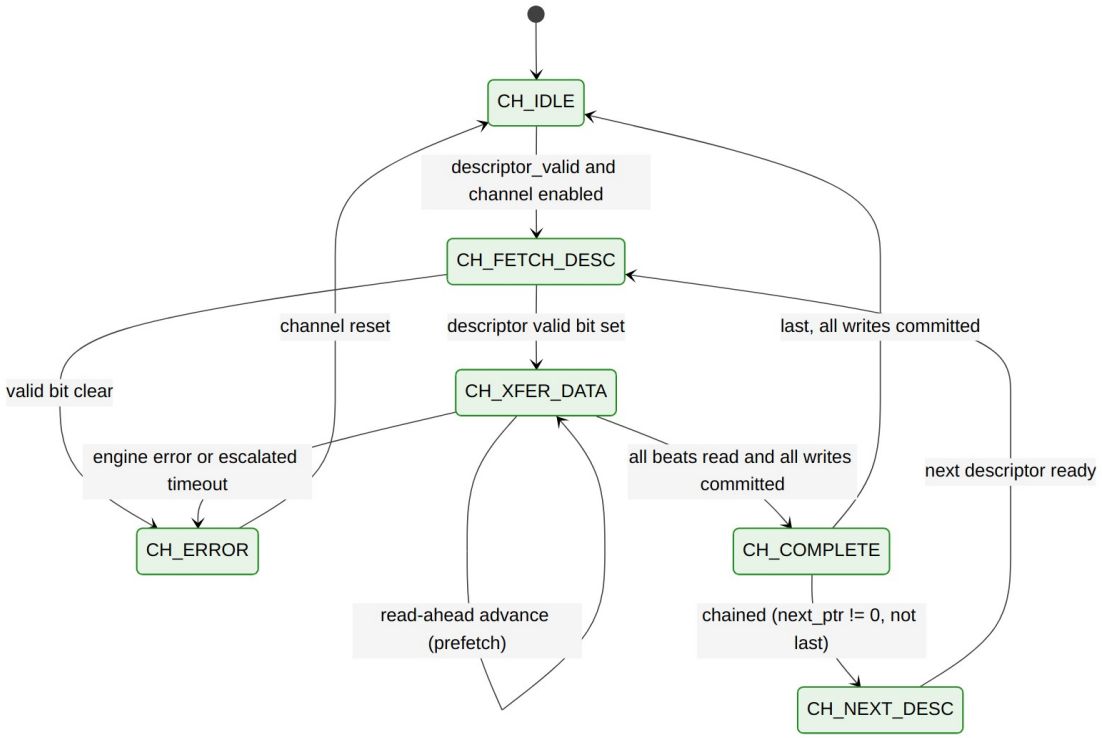
1. **Descriptor Fetch:** When a channel has a descriptor address to fetch. In the default build (`SHARED_ENGINES = 1`) the pick is inside the one shared descriptor engine; with `SHARED_ENGINES = 0` each channel has its own engine and an arbiter merges their AR requests
2. **Read Phase Start:** When channel ready to read source data
3. **Write Phase Start:** When channel ready to write destination

---

11.5 Channel States

Each channel FSM traverses these states:

11.5.1 Figure 3.3.2: Channel FSM State Diagram



Channel FSM State Diagram

Source: 05_channel_fsm.mmd

Channel FSM States

State	Description	Transitions
CH_IDLE	No active transfer	Descriptor ready and channel

State	Description	Transitions
		enabled -> CH_FETCH_DESC
CH_FETCH_DESC	Latch and check the descriptor (one cycle)	Valid bit set -> CH_XFER_DATA; clear -> CH_ERROR
CH_XFER_DATA	Read (source to SRAM) and write (SRAM to destination) run concurrently	All beats read and all writes committed -> CH_COMPLETE
CH_COMPLETE	Descriptor done; check for a chain	Chained -> CH_NEXT_DESC; last, once every write has committed -> CH_IDLE
CH_NEXT_DESC	Wait for the next chained descriptor	Descriptor ready -> CH_FETCH_DESC
CH_ERROR	Sticky fault (engine error, invalid descriptor, escalated timeout)	Channel reset -> CH_IDLE

A disabled channel does not take a descriptor in CH_IDLE. A descriptor already fetched for it waits in the engine's FIFO until the channel is enabled.

——-|————-|————-| | **IDLE** | No active transfer | Kick-off -> FETCH | | **FETCH** | Fetching descriptor | Complete -> PARSE | | **PARSE** | Validating descriptor | Valid -> ARB_READ, Invalid -> ERROR | | **ARB_READ** | Awaiting read grant | Grant -> READ | | **READ** | Reading source data | Complete -> ARB_WRITE | | **ARB_WRITE** | Awaiting write grant | Grant -> WRITE | | **WRITE** | Writing destination | Complete -> CHAIN | | **CHAIN** | Checking for next | Next -> FETCH, Last -> COMPLETE | | **COMPLETE** | Transfer done | IRQ sent -> IDLE | | **ERROR** | Error condition | SW clear -> IDLE |

11.6 SRAM Buffering

Each channel has its own SRAM FIFO (one `sram_controller_unit` per channel), `SRAM_DEPTH` entries deep. The top-level parameter is a per-channel depth, not a pool divided among the channels, and it is fixed at build time. An idle channel's FIFO is not lent to a busy one.

SRAM Buffering by Build

Build	FIFOs	Entries per channel	Total entries
`stream_top_ch`	8	`SRAM_DEPTH`	8 x `SRAM_DEPTH`

Build	FIFOs	Entries per channel	Total entries
8			
stream_top_ch16	16	SRAM_DEPTH	16 x SRAM_DEPTH

The read engine issues a burst for a channel only when that channel's FIFO has room for it. The write engine issues only when the FIFO holds the data. One channel's full or empty FIFO therefore never stalls another channel.

11.7 Error Isolation

Errors are isolated to the affected channel:

- Error in Channel N does not affect the other channels (0..NUM_CHANNELS-1)
- Shared resource errors logged but don't block other channels
- Software must clear error and re-kick to resume

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12 AXI Master Interfaces

12.1 Overview

STREAM includes three AXI4 master interfaces for memory access:

Interface	Purpose	Data Width	Address Width
`m_axi_desc_*`	Descriptor fetch	256 bits	64 bits
`m_axi_rd_*`	Source data read	Parameterizable	64 bits
`m_axi_wr_*`	Destination write	Parameterizable	64 bits

12.2 Descriptor Fetch Master

12.2.1 Signal Summary

Signal	Direction	Width	Description
`m_axi_desc_arid`	Output	ID_WIDTH	Read transaction ID
`m_axi_desc_araddr`	Output	64	Read address
`m_axi_desc_arlen`	Output	8	Burst length (0 = 1 beat)
`m_axi_desc_arsize`	Output	3	Beat size (5 = 32 bytes)
`m_axi_desc_arburst`	Output	2	Burst type (INCR)
`m_axi_desc_arvalid`	Output	1	Address valid
`m_axi_desc_arready`	Input	1	Address ready
`m_axi_desc_rid`	Input	ID_WIDTH	Read data ID
`m_axi_desc_rdata`	Input	256	Read data
`m_axi_desc_rr`	Input	2	Read response

Signal	Direction	Width	Description
esp			
m_axi_desc_rlast	Input	1	Last beat
m_axi_desc_rvalid	Input	1	Read data valid
m_axi_desc_rready	Output	1	Read data ready

12.2.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1 beat	Single 256-bit descriptor
Burst Type	INCR	Fixed increment
Outstanding	1-2	Configurable
Ordering	In-order	Single ID used

12.2.3 Address Alignment

Descriptor addresses must be 32-byte aligned (256-bit boundary).

12.3 Data Read Master

12.3.1 Signal Summary

Signal	Direction	Width	Description
m_axi_rd_arid	Output	ID_WIDTH	Read transaction ID
m_axi_rd_araddr	Output	64	Read address
m_axi_rd_arlen	Output	8	Burst length
m_axi_rd_arsize	Output	3	Beat size
m_axi_rd_arburst	Output	2	Burst type (INCR)

Signal	Direction	Width	Description
m_axi_rd_arvalid	Output	1	Address valid
m_axi_rd_arready	Input	1	Address ready
m_axi_rd_rid	Input	ID_WIDTH	Read data ID
m_axi_rd_rdata	Input	DATA_WIDTH	Read data
m_axi_rd_rresp	Input	2	Read response
m_axi_rd_rlast	Input	1	Last beat
m_axi_rd_rvalid	Input	1	Read data valid
m_axi_rd_rready	Output	1	Read data ready

12.3.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1-256 beats	AXI4 maximum
Burst Type	INCR	Fixed increment
Outstanding	Configurable	4-16 typical
Ordering	In-order	Channel-based ID tagging

12.3.3 Burst Generation

Large transfers are broken into maximum-length bursts:

Transfer Size	Burst Count	Burst Lengths
256 beats	1	256
512 beats	2	256, 256
300 beats	2	256, 44

12.4 Data Write Master

12.4.1 Signal Summary

Signal	Direction	Width	Description
m_axi_wr_awid	Output	ID_WIDTH	Write transaction ID
m_axi_wr_awaddr	Output	64	Write address
m_axi_wr_awlen	Output	8	Burst length
m_axi_wr_awsize	Output	3	Beat size
m_axi_wr_awburst	Output	2	Burst type (INCR)
m_axi_wr_awvalid	Output	1	Address valid
m_axi_wr_awready	Input	1	Address ready
m_axi_wr_wdata	Output	DATA_WIDTH	Write data
m_axi_wr_wstrb	Output	DATA_WIDTH/8	Byte strobes
m_axi_wr_wlast	Output	1	Last beat
m_axi_wr_wvalid	Output	1	Write data valid
m_axi_wr_wready	Input	1	Write data ready
m_axi_wr_bid	Input	ID_WIDTH	Response ID
m_axi_wr_bresp	Input	2	Write response
m_axi_wr_bvalid	Input	1	Response valid
m_axi_wr_bready	Output	1	Response ready

12.4.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1-256 beats	Matches read bursts
Burst Type	INCR	Fixed increment
Outstanding	Configurable	4-16 typical
Write Strobes	All-ones	Aligned transfers only

12.5 Common AXI Attributes

12.5.1 Supported Features

Feature	Descriptor	Read	Write
INCR Burst	Yes	Yes	Yes
WRAP Burst	No	No	No
FIXED Burst	No	No	No
Exclusive	No	No	No
Locked	No	No	No

12.5.2 Error Handling

- SLVERR/DECERR on any channel stops the current transfer
- Error logged in channel status register
- Subsequent descriptors in chain not processed
- Software intervention required to clear and resume

Last Updated: 2026-01-03

13 APB Configuration Interface

13.1 Overview

STREAM provides an APB slave interface for software configuration and control. The interface supports:

- Channel kick-off (write descriptor address to start transfer)
- Status monitoring
- Interrupt control
- Error handling

13.2 Signal Summary

Signal	Direction	Width	Description
`s_apb_paddr`	Input	13	Address bus (8 KB space; reaches monitor block at 0x1000)
`s_apb_psel`	Input	1	Peripheral select
`s_apb_penable`	Input	1	Enable phase
`s_apb_pwrite`	Input	1	Write enable
`s_apb_pwdata`	Input	32	Write data
`s_apb_pstrb`	Input	4	Byte strobes
`s_apb_pready`	Output	1	Ready signal
`s_apb_prdata`	Output	32	Read data
`s_apb_pslverr`	Output	1	Slave error

13.3 Register Map Overview

Two register maps exist: `stream_regs` for `stream_top_ch8` and `stream_regs_16ch` for `stream_top_ch16`. The 16-channel map adds `CH8..CH15_CTRL_{LOW,HIGH}` at 0x040-0x07C, widens the per-channel fields to [15:0], and moves `CH_STATE` to 0x180. The full list of differences is in the MAS register-map chapter. The table below is the 8-channel map.

Offset	Register	Access	Description
0x000	CH0_CTRL_LOW	RW	Channel 0 descriptor address [31:0] (staged)
0x004	CH0_CTRL_HIGH	RW	Channel 0 descriptor address [63:32] (staged)
0x100	GLOBAL_CTRL	RW	Global enable / reset
0x104	GLOBAL_STATUS	RO	SYSTEM_IDLE
0x120	CHANNEL_ENABLE	RW	Per-channel enable [7:0] ([15:0] in the 16-channel map)
0x124	CHANNEL_RESET	RW	Per-channel reset [7:0] ([15:0] in the 16-channel map)
0x128	KICK_ENABLE	RW	Launch staged addresses, one bit per channel
0x140	CHANNEL_IDLE	RO	Per-channel idle [7:0] ([15:0] in the 16-channel map)
0x148	SCHEDULER_IDLE	RO	Scheduler idle
0x150	CH_STATE0_STATE	RO	Channel 0 scheduler state [6:0], one-hot (0x180 in the 16-channel map)
0x170	SCHED_ERROR	RO	Scheduler error status
0x174	AXI_RD_COMPLETE	RO	Read completion, one bit per channel
0x178	AXI_WR_COMPLETE	RO	Write completion, one bit per channel
0x200	SCHED_TIMEOUT_CYCLES	RW	Scheduler timeout cycles
0x204	SCHED_CONFIG	RW	Scheduler enable and packet enables
0x220	DESCENG_CONFIG	RW	Descriptor engine enable / prefetch / FIFO threshold

Offset	Register	Access	Description
0x2A0	AXI_XFER_CONFIG	RW	RD_XFER_BEATS, WR_XFER_BEATS
0x2B0	PERF_CONFIG	RW	Perf profiler enable / mode / clear
0x2C0	OBS_CTRL	RW	Channel observation mux select
0x2D0	PERF_DATA_LOW	RO	Perf capture low word (FIFO head)
0x2D4	PERF_DATA_HIGH	RO	Perf capture high word, same entry; the FIFO pops once BOTH halves are read
0x2D8	PERF_STATUS	RO	EMPTY / FULL / COUNT
0x1000+	Monitor / performance registers	RW/RO	AXI monitor config (0x10C0-0x111F) and per-monitor perf counters (0x1150-0x11F4), in the separate stream_mon_regs regfile

Note: The AXI monitor and performance registers live in a separate `stream_mon_regs` regfile instantiated at offset **0x1000** on the same APB slave. The APB address decode is therefore **13 bits (8 KB)** so those registers are addressable. Detailed monitor register layout is in the MAS (`stream_mas/ch04_registers/register_map.md`).

13.4 Key Registers

13.4.1 Global Control (CTRL) - 0x000

Bits	Field	Access	Description
[0]	ENABLE	RW	Global enable
[1]	SOFT_RESET	RW	Soft reset (self-clearing)
[31:2]	Reserved	-	Reserved

13.4.2 Global Status (STATUS) - 0x004

Bits	Field	Access	Description
[0]	BUSY	RO	Any channel active
[7:1]	Reserved	-	Reserved
[15:8]	CH_ACTIVE	RO	Per-channel active mask
[23:16]	CH_ERROR	RO	Per-channel error mask
[31:24]	Reserved	-	Reserved

13.4.3 Channel Control (CHn_CTRL) - 0x040 + n*0x10

Bits	Field	Access	Description
[31:0]	DESC_ADDR	RW	Descriptor address (kick-off on write)

Writing to CHn_CTRL initiates a transfer: 1. Descriptor address stored 2. Channel transitions from IDLE to FETCH 3. PREADY de-asserted until channel accepts kick-off

13.4.4 Channel Status (CHn_STATUS) - 0x044 + n*0x10

Bits	Field	Access	Description
[3:0]	STATE	RO	Channel FSM state
[4]	IDLE	RO	Channel is idle
[5]	BUSY	RO	Channel is processing
[6]	ERROR	RO	Error condition
[7]	COMPLETE	RO	Last transfer complete
[15:8]	DESC_COUNT	RO	Descriptors processed
[23:16]	ERR_CODE	RO	Error code if ERROR set

Bits	Field	Access	Description
[31:24]	Reserved	-	Reserved

13.4.5 Channel Observation Control (OBS_CTRL) - 0x0C0

Bits	Field	Access	Description
[2:0]	CHANNEL_SEL	RW	Channel index (0-7) for observation read
[31:3]	Reserved	-	Reserved

Software writes the channel index to this register, then reads OBS_FLAGS and OBS_DATA0/DATA1 to observe that channel's internal state.

13.4.6 Channel Observation Flags (OBS_FLAGS) - 0x0C4

Bits	Field	Access	Description
[0]	SCH_ERROR_STICKY	RO	Scheduler error sticky (descriptor invalid, timeout, etc.)
[1]	SCH_TIMEOUT	RO	Scheduler timeout state (waiting on AXI master)
[7:2]	Reserved	-	Reserved
[31:8]	Reserved	-	Reserved

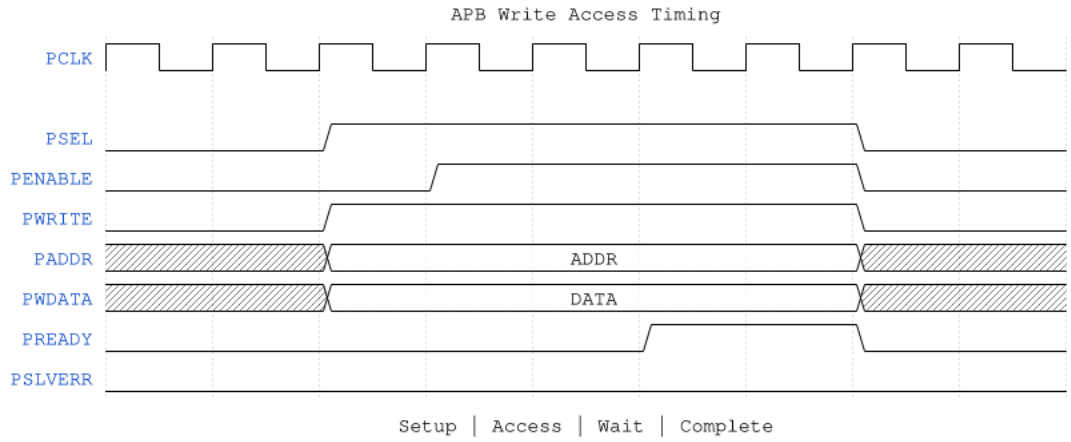
These flags are driven by the scheduler's internal error and timeout monitoring. See `stream_mas/ch02_blocks/04_scheduler.md` for detailed definitions.

13.4.7 Channel Observation Data (OBS_DATA0, OBS_DATA1) - 0x0C8, 0x0CC

Per-channel data such as descriptor state, current beat count, or other diagnostic fields. Exact contents vary by channel state; see the register map audit document for current field assignments.

13.5 Access Timing

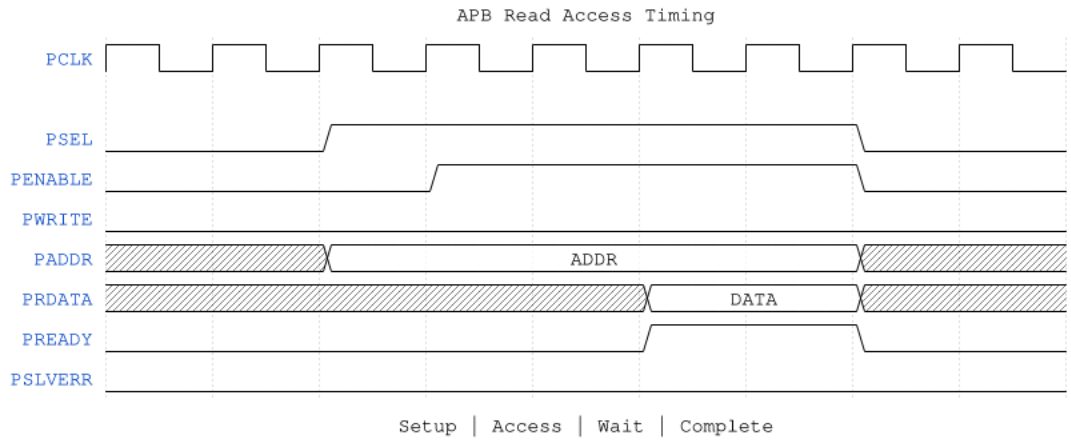
13.5.1 Write Access



APB Write Access Timing

Source: apb_write_access.json

13.5.2 Read Access



APB Read Access Timing

Source: apb_read_access.json

13.6 Kick-off Blocking

When software writes to CHn_CTRL:

- If channel is IDLE: Immediate accept, PREADY asserted
- If channel is BUSY: PREADY held low until channel completes
- Prevents descriptor pointer corruption during active transfer

---

13.7 Launching Channels Together (KICK_ENABLE)

Software stages a 64-bit descriptor address per channel in CHn_CTRL_{LOW,HIGH} (0x000-0x03F, ordinary stored registers) and then launches with **one** write to KICK_ENABLE (0x128). Each KICKn field is a single-pulse: the write emits a one-cycle request and self-clears. Writing an address no longer kicks anything, so staging and launching are separate steps and every channel starts on the same cycle.

This matters when the APB path is reached over a slow transport (a UART-to-APB bridge on an FPGA), where eight separate two-register kicks would serialize channel starts over milliseconds.

Historical note. This replaced two earlier mechanisms: a kick block that snooped the raw APB command stream so that the address write itself kicked (the address was never readable state), and a pair of top-level i_kick_burst_mask / i_kick_burst_addr ports that existed only to work around that latency from harness_csr. **Neither exists in the RTL.** This page documented the i_kick_burst_* ports as a live interface until 2026-09-26; there are no such ports on any module, and KICK_ENABLE needs no port plumbing outside STREAM. See the comment at rtl/top/stream_top_ch8.sv:405-417.

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14 Monitoring Interface

14.1 Overview

STREAM provides a unified Monitor Bus (MonBus) interface for debug and performance monitoring. This interface streams event packets (128-bit) with paired 64-bit side-band timestamps, capturing key operational events from all internal blocks.

14.2 Signal Summary

Signal	Direction	Width	Description
`monbus_pkt_valid`	Output	1	Packet valid
`monbus_pkt_ready`	Input	1	Downstream ready
`monbus_pkt_data`	Output	128	Packet data (protocol-independent monitor packet)
`monbus_ts_data`	Output	64	Side-band timestamp (cycle count at packet capture)

14.3 Packet Format

The 128-bit MonBus packet and 64-bit timestamp are paired atomically and transmitted as a 192-bit group through AXI-Lite skid buffering. See the canonical packet specification at `docs/markdown/rtl-amba/includes/ocelus_package_spec.md` for complete field definitions.

Bits	Field	Description
[63:56]	`SUBSYSTEM_ID`	STREAM subsystem identifier
[55:48]	`EVENT_TYPE`	Event category
[47:40]	`EVENT_CODE`	Specific event within category
[39:32]	`CHANNEL_ID`	Associated channel (0-7)

Bits	Field	Description
[31:0]	PAYLOAD	Event-specific data

14.4 Monitoring Architecture

14.4.1 Descriptor Path Monitoring (Both Ends)

STREAM monitors the descriptor fetch path at **both ends** for observability:

1. **Descriptor Fetch Master (Agent ID 8):** Monitors the AXI4 master that fetches descriptors from system memory
2. **Descriptor Consume Side (Agent ID 48 + channel):** Monitors when the scheduler consumes the descriptor for processing

This dual-end monitoring allows software to attribute descriptor path stalls to either fetch delay (master side) or consume delay (slave side), enabling faster root-cause analysis.

14.4.2 Global Monitor Enable Methodology

All internal monitors are gated by: - **Global SV parameter:** `USE_MONITOR` (default: enabled for analysis, disabled for area-optimized builds) - **Per-port SV parameters:** Individual monitor enable for each monitored port - **APB runtime configuration:** Masks for enabling/disabling packet types at run time

---

14.5 Event Types

STREAM uses the unified MonBus event categorization defined in `docs/markdown/rtl-amba/includes/ocelus_package_spec.md`: - **Error events (Type 0x0):** Protocol violations, timeouts, invalid descriptors - **Completion events (Type 0x1):** Successful transfer completion, descriptor processed - **Performance events (Type 0x4):** Latency metrics, throughput data - **Protocol-specific events:** AXI address-match, debug traces

All STREAM events use **Unit ID** = **1** and channel-specific **Agent IDs** for the source block (descriptor engine, scheduler, AXI monitor, etc.).

Do not re-state the 128-bit packet layout here — refer to the canonical specification for field definitions and bit assignments.

14.6 Event Configuration

14.6.1 Configurable Events

Events can be enabled/disabled via APB registers:

Register	Bit	Event Category
MONBUS_CFG	[0]	Transfer events
MONBUS_CFG	[1]	Error events
MONBUS_CFG	[2]	Performance events
MONBUS_CFG	[7:4]	Channel mask

14.6.2 Event Priority

When multiple events occur simultaneously:

1. Error events (highest priority)
2. Transfer completion events
3. Transfer start events
4. Performance events (lowest priority)

14.7 Bulk-Trace Compression

The MonBus output can be compressed before it leaves STREAM. Compression is a build option with a runtime enable on top of it:

Control	Kind	Meaning
USE_MON_COMPRESSION	parameter	Builds the compressor into the monbus group. 0 compiles it out entirely.
USE_MON_HALFBEAT	parameter	Packs two 30-bit slots per 64-bit beat. Only meaningful with USE_MON_COMPRESSION=1.
COMPRESS_EN	register field	Runtime enable. Only meaningful when the compressor was built.

Both parameters default to 1 via `CFG_USE_MON_COMPRESSION` / `CFG_USE_MON_HALFBEAT`. A runtime `COMPRESS_EN` on a build that compiled the compressor out is silently inert, so read the build parameter before concluding anything from the enable bit.

What compression changes on the wire. Uncompressed, each 128-bit packet occupies three 64-bit beats (`{tag, timestamp}`, `packet[127:64]`, `packet[63:0]`). Compressed, the group emits variable compressor slots with the tag in bits [63:60]. The two encodings are not interchangeable, which has one important consequence for integrators:

- A **capture memory** is the destination for compressed traffic. The host reads it back and decodes it with the reference model.
- A **tally** cannot consume compressed records. It reassembles raw 3-beat records with a mod-3 beat counter and has no compressor-slot decoder, so pointing compressed traffic at a tally silently produces nothing.

A useful sanity check when reading a capture back: uncompressed data lands at exactly 3.00 slots per packet. Any measurement at 3.00 is raw, whatever the enable bits say.

Measured (cosim, 2026-09-08). With the compressor built and half-beat packing on, a monitors-on DMA workload captured 36 slots for 32 decoded packets: **1.12 slots per packet, 62.5% smaller** than the raw 3-beat encoding. The same workload with the compressor compiled out produced 96 slots for the same 32 packets – exactly 3.00, the raw ratio. Both decoded identically, so the difference is encoding alone.

Implementation is documented globally, not here. The encoder, its slot format, the half-beat packing and the group that carries them are specified in:

- `docs/markdown/rtl-amba/monitor/monbus_compressor.md` – the encoder and slot format
- `docs/markdown/rtl-amba/monitor/monbus_halfbeat_packer.md` – half-beat packing
- `docs/markdown/rtl-amba/monitor/monbus_group_core.md` – the group, its FIFOs and the beat layouts for both modes

Those are the canonical descriptions. This section states only what a STREAM integrator must decide: whether to build the compressor, and where the records are allowed to land.

14.8 Backpressure Handling

14.8.1 Flow Control

- Events generated when `monbus_pkt_ready` is asserted
- If `monbus_pkt_ready` is low, events are queued in internal FIFO

- FIFO depth: 64 entries (configurable)
- Overflow behavior: Oldest events discarded, overflow counter incremented

14.8.2 Recommendations

- Connect MonBus to FIFO with sufficient depth
- Monitor overflow counter in status register
- Size downstream FIFO based on expected event rate

14.9 Typical MonBus Downstream

```
STREAM MonBus → MonBus FIFO → MonBus Arbiter → System Trace
                 (64 deep)    (multi-source)    Interface
```

14.10 Example Event Sequence

A single-descriptor transfer generates these events:

```
Time    Event
----    -----
T0      KICKOFF     - Software kicks off channel
T1      DESC_FETCH  - Descriptor fetch initiated
T10     DESC_VALID  - Descriptor parsed successfully
T12     ARB_WAIT    - Waiting for data path grant (if applicable)
T15     READ_START  - AXI read burst initiated
T75     READ_DONE   - All source data read
T80     WRITE_START - AXI write burst initiated
T140    WRITE_DONE  - All data written, responses received
T141    COMPLETE    - Channel complete, back to IDLE
```

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15 Throughput Targets

15.1 Theoretical Maximum

15.1.1 Per-Interface Bandwidth

Interface	Data Width	Frequency	Bandwidth
Descriptor Fetch	256 bits	200 MHz	6.4 GB/s
Data Read	512 bits	200 MHz	12.8 GB/s
Data Write	512 bits	200 MHz	12.8 GB/s

15.1.2 Aggregate Bandwidth

- **Bidirectional:** 25.6 GB/s (simultaneous read + write)
- **Descriptor Overhead:** ~0.1% for large transfers

15.2 Practical Throughput

15.2.1 Single-Channel Performance

Scenario	Expected Efficiency	Effective Bandwidth
Large sequential (>1MB)	>95%	>12.0 GB/s
Medium sequential (64KB-1MB)	85-95%	10.8-12.0 GB/s
Small sequential (4KB-64KB)	70-85%	9.0-10.8 GB/s
Very small (<4KB)	40-70%	5.0-9.0 GB/s

15.2.2 Multi-Channel Performance

Active Channels	Per-Channel BW	Aggregate BW
1	12.8 GB/s	12.8 GB/s
2	6.4 GB/s	12.8 GB/s
4	3.2 GB/s	12.8 GB/s
8	1.6 GB/s	12.8 GB/s

Note: Aggregate limited by shared AXI masters. Individual channel throughput scales inversely with active channel count.

15.2.3 Measured on the Genesys 2: Shared and Per-Channel Descriptor Engines

Source: the two bitstreams of the Resource Estimates chapter, `stream_perf_genesys2_8ch_pce.bit` (`SHARED_ENGINES = 0`) and `stream_perf_genesys2_8ch.bit` (`SHARED_ENGINES = 1`): 8 channels, 128-bit data, 100 MHz. Both ran back to back on 2026-10-17 on the same board under the same host sweep (`host_characterize.py`): 1, 2, 4, 8 and 16 descriptors per channel, 1 to 8 active channels, 1 MiB per channel, and no added memory latency. That is 40 configurations. The records are `projects/fpga-systems/Genesys2/dma-ip/stream/reports/perf/json/stream_sweep_20261017_shared_engines0.json` and `..._shared_engines1.json`.

Genesys 2 Board Results, SHARED_ENGINES 0 and 1

Measure	Per-channel engines	Shared engine
Configurations passed	40 of 40	40 of 40
Data rate, first read to last read and first write to last write	1525.9 MiB/s	1525.9 MiB/s
Theoretical peak, 16 bytes x 100 MHz	1525.9 MiB/s (1600 MB/s)	1525.9 MiB/s (1600 MB/s)
Lowest end-to-end rate, kick to done	1524.8 MiB/s	1524.8 MiB/s
Total cycles, shared minus per-channel		-1 in every configuration

Both builds move one beat per cycle from the first beat to the last on both AXI masters, which is the line rate. The data-phase cycle counts are identical in all 40 configurations. The only difference is that every run with the shared engine finishes exactly one cycle sooner, a fixed start-up difference in the descriptor path. Sharing the descriptor engine costs no throughput.

15.3 Throughput Limiting Factors

15.3.1 Memory System Factors

Factor	Impact	Mitigation
Memory latency	Reduces efficiency for small transfers	Use deeper outstanding
Memory bandwidth	Hard limit	Match STREAM to memory capability
Interconnect contention	Variable impact	Priority configuration

15.3.2 STREAM Internal Factors

Factor	Impact	Mitigation
Descriptor fetch overhead	Fixed per descriptor	Use longer transfers
Per-descriptor boundary bubble	~13 cyc/descriptor (chains)	Read-ahead prefetch (default on) — eliminates it
SRAM depth	Limits outstanding	Configure adequate depth
Arbitration overhead	Multi-channel penalty	Reduce active channels
Burst splitting	4KB boundary crossings	Align transfers

15.4 Performance Targets

15.4.1 Design Targets

Metric	Target	Condition
Peak throughput	12.8 GB/s	Single direction, ideal
Sustained throughput	>11.0 GB/s	Large transfers, single channel
Multi-channel aggregate	>10.0 GB/s	8 channels active
Small transfer efficiency	>50%	4KB transfers

15.4.2 Verification Criteria

Scenario	Pass Criteria
1MB sequential read	>95% efficiency
1MB sequential write	>95% efficiency
64KB scatter (4x16KB)	>80% efficiency
4KB × 100 descriptors	>60% efficiency

15.5 Descriptor Chain Efficiency

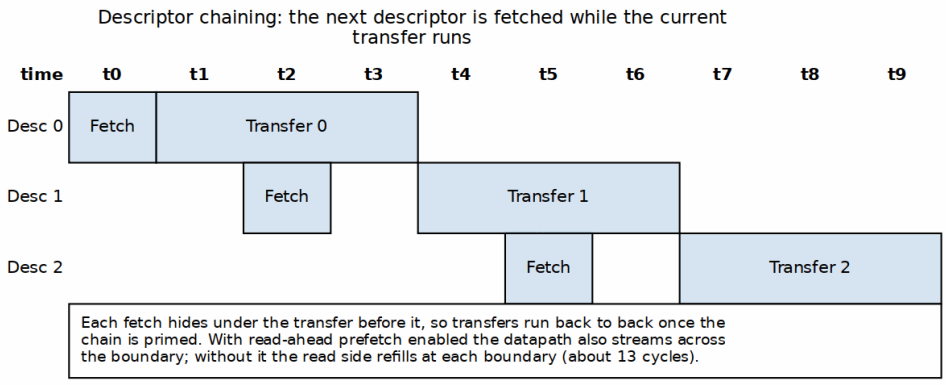
15.5.1 Descriptor Overhead

Transfer Size	Descriptor Fetches	Overhead
64 KB	1	0.05%
256 KB	1	0.01%
1 MB	1	<0.01%
4 KB (100 chained)	100	0.5%

15.5.2 Chaining Efficiency

For chained descriptors, next descriptor fetch can overlap with current transfer:

15.5.3 Figure 5.1.1: Descriptor Chain Fetch Overlap



Descriptor Chain Fetch Overlap

This pipelining minimizes chaining overhead for continuous transfers.

15.5.4 Read-Ahead Descriptor Prefetch (Cross-Descriptor Streaming)

Descriptor fetch overlaps the current transfer (above), but the datapath itself must also stream across the boundary. Without prefetch the read side stalls at each descriptor boundary while the scheduler FSM advances and the read pipeline (AR -> R) refills — a per-descriptor **bubble** of ~13 cycles that dominates on short-descriptor chains. Read-ahead descriptor prefetch (runtime `SCHED_CONFIG.RD_PREFETCH_EN`, default **on**; see MAS Section 2.4) advances the read side to descriptor N+1 while the write side still drains N, so the read stream never goes cold. The per-descriptor bubble drops to **zero** — chain throughput becomes independent of descriptor count, with only a one-time fill/drain of overhead.

Measured on silicon (Nexys A7, 100 MHz, one bitstream, prefetch on vs off):

Descriptor size	Prefetch OFF (lockstep)	Prefetch ON
64 beats (16-desc chain)	76.1% datapath util	95.3%
>= 4096 beats	~99%	99-100%

Applies to chained **legacy** descriptors. EXT (row/col-major) descriptors run lockstep, but a single row-major descriptor is internally bubble-free (its contiguous runs stream through the address-generator base FIFO — measured 99.7% util). Transpose access patterns (col-major on either side) are single-beat and latency-bound by the memory system, not by a scheduler bubble.

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16 Latency Characteristics

16.1 Latency Components

16.1.1 End-to-End Latency Breakdown

For a single-descriptor transfer (kick-off to completion):

Phase	Typical Cycles	Notes
APB write processing	2-3	APB transaction
Descriptor fetch	10-50	Memory latency dependent
Descriptor parse	2-4	Fixed pipeline
Arbitration wait	0-100+	Depends on contention
Read phase setup	2-3	AXI AR channel
Read data transfer	N + 10-50	N beats + memory latency
Write phase setup	2-3	AXI AW channel
Write data transfer	N + 5-20	N beats + response latency
Completion handling	2-4	Status update, IRQ

16.1.2 Total Latency Formula

```
Latency = T_apb + T_desc_fetch + T_parse + T_arb + T_read + T_write +
T_complete

Where:
  T_apb        = 3 cycles (fixed)
  T_desc_fetch = Memory_latency + 10 cycles
  T_parse      = 3 cycles (fixed)
  T_arb        = 0 to N (contention dependent)
  T_read       = Memory_latency + (Transfer_length / Beats_per_burst)
* Burst_overhead + Transfer_length
  T_write      = Memory_latency + Transfer_length + Response_latency
  T_complete   = 3 cycles (fixed)
```

16.2 First-Byte Latency

Time from kick-off to first data beat arriving at destination:

Component	Cycles	Cumulative
APB processing	3	3
Descriptor fetch	30	33
Parse + setup	5	38
Read AR accepted	10	48
First read data	20	68
Write AW + first W	10	78
Total	**~80**	-

At 200 MHz: ~400 ns first-byte latency (typical)

16.3 Latency by Transfer Size

16.3.1 Small Transfers

Transfer Size	Total Latency	Efficiency
64 bytes (1 beat)	~100 cycles	1%
256 bytes (4 beats)	~110 cycles	4%
1 KB (16 beats)	~130 cycles	12%
4 KB (64 beats)	~180 cycles	35%

16.3.2 Large Transfers

Transfer Size	Total Latency	Efficiency
16 KB (256 beats)	~400 cycles	64%
64 KB (1024 beats)	~1200 cycles	85%
256 KB (4096 beats)	~4400 cycles	93%
1 MB (16384 beats)	~17000 cycles	96%

16.4 Latency Variability

16.4.1 Sources of Variability

Source	Impact	Range
Memory system load	High	10-100+ cycles
AXI interconnect	Medium	5-50 cycles
Channel arbitration	Medium	0-100+ cycles
Descriptor chaining	Low	2-5 cycles

16.4.2 Worst-Case Latency

Conditions for worst-case latency:

1. All channels active (maximum arbitration delay: 8 or 16 channels, by top)
2. Memory system fully loaded (maximum memory latency)
3. Small transfer size (maximum overhead ratio)
4. Low priority descriptor (maximum wait time)

Under these conditions, first-byte latency could exceed 1000 cycles.

16.5 Latency Optimization

16.5.1 Software Recommendations

Recommendation	Benefit
Use larger transfers	Better efficiency
Minimize active channels	Reduce arbitration
Use higher priority for latency-sensitive	Earlier service
Pre-stage descriptors in cache	Faster fetch

16.5.2 Hardware Configuration

Setting	Effect
Increase outstanding limit	Hide memory latency
Enable descriptor prefetch	Reduce chaining overhead
Configure larger SRAM	Support longer bursts

16.6 Interrupt Latency

Time from transfer completion to interrupt assertion:

Phase	Cycles
Last write response	0
Completion FSM	2
Status register update	1
Interrupt assertion	1
Total	**4 cycles**

At 200 MHz: 20 ns from completion to interrupt.

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17 Resource Estimates

17.1 Measured Core Area

Out-of-context Vivado synthesis of `stream_core` at 256-bit data, with monitors off and a 100 MHz clock (2026-10-17):

Measured stream_core Area and Timing

Build	LUTs	Flops	Worst slack
8 channels, one descriptor engine per channel (`SHARED_ENGINES = 0`)	25.8k	14.9k	+3.24 ns
8 channels, shared descriptor engine (default)	20.7k	11.7k	+2.50 ns
16 channels, shared descriptor engine	36.8k	19.4k	+1.79 ns

Sharing the descriptor engine removes about a fifth of the 8-channel core's LUTs and flops: the per-channel AR issue logic, the address-range comparators, the 256-bit holding registers and the MonBus reporters. The 16-channel build costs about 1.8 times the 8-channel build and still closes at 100 MHz.

17.2 Measured Board Build (Genesys 2)

Source: two placed-and-routed Genesys 2 builds made on 2026-10-17 from the same RTL, differing only in `SHARED_ENGINES`. The bitstreams are `stream_perf_genesys2_8ch_pce.bit` (`SHARED_ENGINES = 0`) and `stream_perf_genesys2_8ch.bit` (`SHARED_ENGINES = 1`). Both builds use 8 channels, 128-bit data, 256 entries per channel, 8 outstanding reads and writes, AXI monitors off and `GEN_MON = 0`, at 100 MHz on the `xc7k325tffg900-2` with Vivado 2025.1. The STREAM rows are the `u_stream` instance of the hierarchical post-route report; the whole-design rows include the characterization harness.

Post-Route Utilization, SHARED_ENGINES 0 and 1

Resource	Per-channel engines	Shared engine	Change
Whole design, Slice LUTs	63,642	62,111	-1,531 (-2.4 %)
Whole design, Slice registers	56,536	54,111	-2,425 (-4.3 %)
STREAM (`u_stream`), LUTs	30,781	29,247	-1,534 (-5.0 %)
STREAM (`u_stream`), flip-flops	23,079	20,654	-2,425 (-10.5 %)
Descriptor engines, LUTs / flip-flops	4,974 / 4,120 (8 engines)	3,825 / 1,725 (1 engine)	-1,149 / -2,395
Block RAM tiles	26	26	0
DSP	64	64	0
Worst setup slack	+0.612 ns	+0.377 ns	

The board saving is smaller than the out-of-context figure above for two reasons. The board build has `GEN_MON = 0`, so the per-channel MonBus reporters that sharing removes are not built in either case. And the shared engine keeps a descriptor FIFO and an extended-descriptor FIFO per channel; together they are 1,840 LUTs of distributed RAM, almost half the engine. Block RAM and DSP are unchanged because the descriptor engine uses neither. Both builds meet timing with no failing endpoint.

17.3 FPGA Resource Summary (Estimates)

The estimates below predate the measurements above and are kept for the per-block breakdown. Typical configuration (8 channels, 512-bit data width, 4096-entry SRAM per channel):

Resource	Estimate	Notes
LUTs	15,000 - 25,000	Logic and control

Resource	Estimate	Notes
Registers	20,000 - 30,000	Pipeline and state
Block RAM	40 - 60	SRAM buffer
DSP	0	No DSP usage

17.4 Resource Breakdown by Block

17.4.1 APB Configuration Slave

Resource	Estimate
LUTs	500 - 800
Registers	400 - 600
BRAM	0

17.4.2 Descriptor Engine

One shared engine for every channel in the default build.

Resource	Estimate
LUTs	1,500 - 2,500
Registers	1,000 - 1,500
BRAM	1 (descriptor FIFO)

17.4.3 Scheduler (8 channels)

Resource	Estimate
LUTs	3,000 - 5,000
Registers	4,000 - 6,000
BRAM	0

17.4.4 Channel Arbiter

Resource	Estimate
LUTs	500 - 1,000
Registers	200 - 400
BRAM	0

17.4.5 AXI Read Engine

Resource	Estimate
LUTs	2,000 - 3,500
Registers	3,000 - 4,500
BRAM	0

17.4.6 SRAM Buffer

Resource	Estimate
LUTs	100 - 200
Registers	100 - 200
BRAM	32 - 64 (depth/width dependent)

17.4.7 AXI Write Engine

Resource	Estimate
LUTs	2,000 - 3,500
Registers	3,000 - 4,500
BRAM	0

17.4.8 MonBus Reporter

Resource	Estimate
LUTs	800 - 1,200
Registers	600 - 900
BRAM	1 (event FIFO)

17.5 SRAM Sizing

SRAM buffer size depends on configuration:

Configuration	BRAM (18Kb)	BRAM (36Kb)
512b x 1024	16	8
512b x 2048	32	16
512b x 4096	64	32
256b x 4096	32	16
128b x 4096	16	8

17.6 Scaling with Parameters

17.6.1 Channel Count Impact

Channels	LUT Delta	Register Delta
1	Base	Base
2	+500	+800
4	+1,500	+2,400
8	+3,500	+5,600
16	measured: see Measured Core Area	

17.6.2 Data Width Impact

Data Width	LUT Delta	Register Delta	BRAM Delta
128 bits	-30%	-40%	-75%
256 bits	-15%	-20%	-50%
512 bits	Base	Base	Base

17.7 ASIC Estimates

For 28nm technology node (typical):

Metric	Estimate
Gate Count	200K - 350K gates
Area	0.2 - 0.4 mm2
SRAM Area	0.1 - 0.2 mm2 (4096x512)
Total Area	0.3 - 0.6 mm2

17.7.1 Power Estimates

Condition	Power
Idle (clock gated)	<1 mW
Idle (clock running)	10 - 20 mW
Single channel active	50 - 100 mW

Condition	Power
All channels active	150 - 300 mW

17.8 Timing Estimates

17.8.1 Target Frequencies

Technology	Target Fmax
Xilinx Ultrascale+	250 MHz
Agilex	250 MHz
28nm ASIC	400 MHz
16nm ASIC	600 MHz

17.8.2 Critical Paths

Path	Concern Level	Mitigation
AXI data path	Low	Fully pipelined
SRAM read/write	Low	Registered outputs
Arbiter decision	Medium	Priority encoder depth
Descriptor parse	Low	Simple combinational

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18 System Requirements

18.1 Interface Requirements

18.1.1 APB Interface

Requirement	Specification
Protocol	APB3 or APB4
Data Width	32 bits
Address Width	12 bits minimum
Timing	Synchronous to `aclk`

18.1.2 AXI Interfaces

Requirement	Descriptor	Read Data	Write Data
Protocol	AXI4	AXI4	AXI4
Data Width	256 bits	Parameterizable	Parameterizable
Address Width	64 bits	64 bits	64 bits
ID Width	4 bits min	4 bits min	4 bits min
Outstanding	2	Configurable	Configurable

18.1.3 MonBus Interface

Requirement	Specification
Data Width	64 bits
Protocol	Valid/Ready streaming
Backpressure	Required (must handle ready deassertion)

18.2 Memory Requirements

18.2.1 Descriptor Memory

Requirement	Specification
Alignment	32-byte aligned

Requirement	Specification
Access	Read access via descriptor AXI master
Coherency	Software responsibility
Latency	<100 cycles recommended

18.2.2 Data Memory

Requirement	Specification
Alignment	Per data width (64-byte for 512-bit)
Access	Read via read master, Write via write master
Bandwidth	Sufficient for target throughput

18.3 System Bandwidth Allocation

18.3.1 AXI Interconnect Requirements

To achieve maximum STREAM throughput:

AXI Master	Required Bandwidth	Priority
Descriptor Fetch	0.1 GB/s	Medium
Data Read	12.8 GB/s	High
Data Write	12.8 GB/s	High

18.3.2 Interconnect Recommendations

- Configure STREAM AXI masters with high priority
- Ensure sufficient outstanding transaction support
- Consider dedicated paths for high-bandwidth DMA

18.4 Interrupt Requirements

18.4.1 Interrupt Controller

Requirement	Specification
Interrupt Lines	8 (per-channel) or 1 (combined)

Requirement	Specification
Type	Level-sensitive, active-high
Latency	<10 cycles from assertion to controller

18.4.2 Interrupt Handling

- Software must read `IRQ_STATUS` to identify source
- Write-1-to-clear to acknowledge interrupts
- Channel remains blocked until error cleared (if applicable)

18.5 Address Space Requirements

18.5.1 STREAM Register Space

Requirement	Specification
Base Address	System-defined
Size	4 KB (0x1000)
Alignment	4 KB aligned

18.5.2 Memory Map Constraints

- Descriptor addresses must be accessible via descriptor AXI master
- Source addresses must be accessible via read AXI master
- Destination addresses must be accessible via write AXI master
- All addresses must respect configured address range limits

18.6 Software Requirements

18.6.1 Driver Requirements

Requirement	Description
Descriptor Preparation	Build descriptors in coherent memory
Cache Management	Flush descriptor cache before kick-off
Interrupt Handling	Service interrupts promptly

Requirement	Description
Error Recovery	Clear error status, optionally retry

18.6.2 Programming Sequence

1. Verify STREAM is enabled (`CTRL.ENABLE = 1`)
2. Check target channel is idle (`CHn_STATUS.IDLE = 1`)
3. Prepare descriptor chain in memory
4. Flush cache (if applicable)
5. Write first descriptor address to `CHn_CTRL`
6. Wait for interrupt or poll `CHn_STATUS`

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19 Clocking and Reset

19.1 Clock Requirements

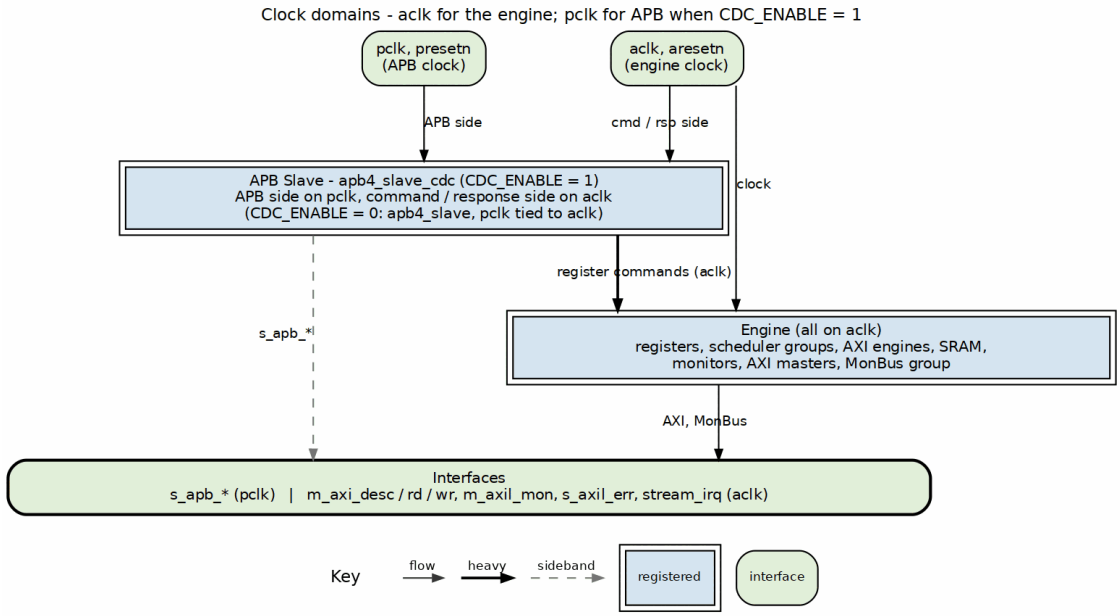
19.1.1 Primary Clock (aclk)

Parameter	Requirement
Frequency	100 - 400 MHz (target dependent)
Duty Cycle	40% - 60%
Jitter	<100 ps peak-to-peak
Domain	All logic except the APB slave side (see below)

19.1.2 Clock Relationship

Everything is synchronous to `aclk` except the APB slave interface. With `CDC_ENABLE = 1` (the `stream_top_ch8` default) `s_apb_*` runs on `pclk` / `presetn` and `apb4_slave_cdc` crosses into `aclk`; with `CDC_ENABLE = 0` `apb4_slave` is used and the APB interface runs on `aclk`:

19.1.3 Figure 6.2.1: Clock Domains



Clock Domains

Source: ch06_clock_distribution.dot

[INTEG-1] The AXI, AXI-Lite and interrupt interfaces must be synchronous to `aclk`. The APB interface may use its own `pclk` when `CDC_ENABLE = 1`; any other clock domain crossing is the system integrator's responsibility.

19.2 Reset Requirements

19.2.1 Reset Signal (aresetn)

Parameter	Requirement
Polarity	Active-low
Type	Asynchronous assert, synchronous deassert
Minimum Pulse	2 clock cycles
Glitch Filter	Not required (external synchronizer assumed)

19.2.2 Reset Sequence

Reset is asserted asynchronously and de-asserted synchronously:

1. `aresetn` goes low (asynchronous assertion)
2. Reset remains active for minimum 2 clock cycles
3. `aresetn` goes high on rising edge of `aclk` (synchronous release)
4. All blocks begin normal operation

19.2.3 Reset Behavior

Block	Reset State
APB Config	All registers to default
Scheduler	All channels IDLE
Descriptor Engine	FSM to IDLE, FIFOs empty
AXI Engines	No outstanding transactions
SRAM	Contents undefined
MonBus	FIFO empty, no output

19.3 Power Management

19.3.1 Clock Gating Support

STREAM supports external clock gating when idle:

Signal	Purpose
`all_channels_idle`	Indicates safe to gate clock

Clock Gating Sequence: 1. Check `STATUS.BUSY = 0` 2. Disable clock via external gate 3. Re-enable clock before any APB access

19.3.2 Low Power Recommendations

Technique	Implementation
Clock gating	Gate `aclk` when `all_channels_idle`
Power gating	Not recommended (state loss)
Voltage scaling	Supported if timing met

19.4 Timing Constraints

19.4.1 Clock Period Constraints

```
# Example Vivado XDC constraints
create_clock -period 5.000 -name aclk [get_ports aclk]

# APB interface (same clock)
set_input_delay -clock aclk 1.0 [get_ports s_apb_*]
set_output_delay -clock aclk 1.0 [get_ports s_apb_pready]
set_output_delay -clock aclk 1.0 [get_ports s_apb_prdata*]
set_output_delay -clock aclk 1.0 [get_ports s_apb_pslverr]

# AXI interfaces (same clock)
set_input_delay -clock aclk 1.0 [get_ports m_axi_*ready]
set_input_delay -clock aclk 1.0 [get_ports m_axi_*rdata*]
set_input_delay -clock aclk 1.0 [get_ports m_axi_*rresp*]
set_input_delay -clock aclk 1.0 [get_ports m_axi_*bresp*]
set_output_delay -clock aclk 1.0 [get_ports m_axi_*valid]
set_output_delay -clock aclk 1.0 [get_ports m_axi_*addr*]
set_output_delay -clock aclk 1.0 [get_ports m_axi_*data*]
```

19.4.2 False Path Constraints

```
# Reset is asynchronous
set_false_path -from [get_ports aresetn]

# Configuration registers to datapath (multi-cycle)
set_multicycle_path 2 -setup -from [get_cells *cfg_*]
set_multicycle_path 1 -hold -from [get_cells *cfg_*]
```

19.5 CDC Considerations

19.5.1 Single Clock Domain

STREAM operates in a single clock domain. All clock domain crossing must be handled externally:

Interface	CDC Responsibility
APB Slave	External if APB on different clock
AXI Masters	External if AXI interconnect on different clock
MonBus	External if consumer on different clock

19.5.2 CDC Recommendations

For systems requiring CDC:

1. Use async FIFO for MonBus output
2. Use AXI clock converter IP for AXI interfaces
3. Use APB bridge with clock crossing for APB

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20 Verification Strategy

20.1 Verification Approach

20.1.1 Multi-Level Verification

STREAM verification follows a hierarchical approach:

Level	Focus	Coverage Target
Block	Individual module function	100% line/toggle
Integration	Block-to-block interfaces	95% functional
System	End-to-end operation	90% use cases

20.2 Block-Level Verification

20.2.1 Test Categories

Block	Key Test Scenarios
APB Config	Register access, kick-off timing, error response
Descriptor Engine	Fetch, parse, chain, error handling
Scheduler	State transitions, multi-channel coordination
Channel Arbiter	Priority, fairness, starvation prevention
AXI Read Engine	Burst generation, error handling, backpressure
AXI Write Engine	Burst generation, response handling
SRAM Buffer	Read/write concurrent, full/empty
MonBus Reporter	Event generation, backpressure,

Block	Key Test Scenarios
	overflow

20.2.2 Coverage Metrics

Metric	Target
Line coverage	100%
Toggle coverage	95%
FSM state coverage	100%
FSM transition coverage	100%
Parameter coverage	All supported values

20.3 Integration-Level Verification

20.3.1 Interface Verification

Interface	Test Focus
Kick registers to Descriptor Engine	Kick-off handshake, blocking behavior
Descriptor Engine to Scheduler	Descriptor passing, backpressure
Scheduler to AXI Engines	Grant/request, data flow
AXI Engines to SRAM	Concurrent access, pointer management
All blocks to MonBus	Event generation, arbitration

20.3.2 Multi-Channel Scenarios

Scenario	Description
Sequential	Channels operate one at a time
Concurrent	All channels active simultaneously (8 on `stream_top_ch8`, 16 on `stream_top_ch16`)
Priority	High priority preempts low priority
Fairness	Equal priority channels share fairly
Error isolation	Error in one channel, others continue

20.4 System-Level Verification

20.4.1 End-to-End Test Cases

Test Case	Description
Basic Transfer	Single descriptor, single channel
Chained Transfer	Multiple descriptors, single channel
Multi-Channel	Multiple channels, independent transfers
Scatter	Single source to multiple destinations
Gather	Multiple sources to single destination
Maximum Throughput	Sustained bandwidth test
Error Injection	AXI errors, invalid descriptors
Stress	Random traffic, long duration

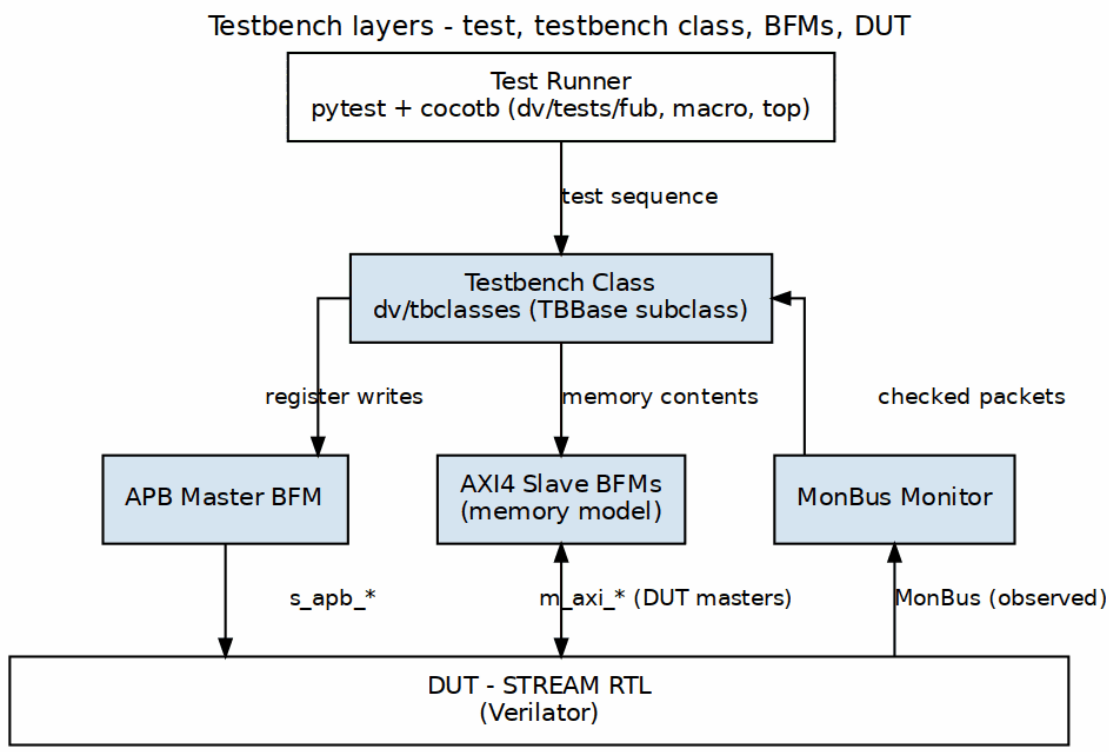
20.4.2 Performance Verification

Metric	Verification Method
Throughput	Measure bytes/cycle over large transfer
Latency	Measure kick-off to completion cycles
Efficiency	Compare actual vs theoretical maximum

20.5 Testbench Architecture

20.5.1 CocoTB-Based Verification

20.5.2 Figure 6.3.1: Testbench Architecture



Testbench Architecture

Source: ch06_testbench_stack.dot

20.5.3 Key Testbench Components

Component	Purpose
APB Master BFM	Drive configuration interface
AXI Slave BFM	Model memory system
MonBus Monitor	Capture and verify events
Scoreboard	Verify data integrity
Coverage Collector	Track functional coverage

20.6 Verification Environment

20.6.1 Simulator Support

Simulator	Status
Verilator	Primary (open-source)
Icarus Verilog	Secondary
Commercial (VCS, Questa)	Supported

20.6.2 Test Execution

```
# Run all STREAM tests
pytest projects/components/dma-ip/stream/dv/tests/ -v

# Run specific test category
pytest projects/components/dma-ip/stream/dv/tests/fub/ -v
pytest projects/components/dma-ip/stream/dv/tests/macro/ -v
pytest projects/components/dma-ip/stream/dv/tests/top/ -v

# Generate coverage report
pytest projects/components/dma-ip/stream/dv/tests/ --cov-report=html
```

20.7 Formal Verification

20.7.1 Properties for Formal

Category	Properties
Protocol	AXI handshake rules, APB timing
Liveness	No deadlock in arbitration
Safety	No data corruption, proper error handling
Functional	Descriptor chaining correctness

20.7.2 Formal Tool Usage

Recommended for: - AXI protocol compliance - Arbiter deadlock freedom - FSM reachability analysis

---

20.8 Sign-Off Criteria

20.8.1 Block-Level Sign-Off

- ☐ 100% line coverage
- ☐ 95% toggle coverage
- ☐ All FSM states exercised
- ☐ All FSM transitions exercised
- ☐ No simulation errors
- ☐ No lint warnings

20.8.2 Integration Sign-Off

- ☐ All interface scenarios passed
- ☐ Multi-channel concurrency verified
- ☐ Error handling verified
- ☐ Performance targets met

20.8.3 System Sign-Off

- ☐ All end-to-end test cases passed
- ☐ Throughput targets achieved
- ☐ Latency targets achieved
- ☐ 72-hour stress test passed

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21 STREAM Specification Index

Version: 0.100 **Date:** 2026-10-20 **Purpose:** Complete technical specification for STREAM subsystem

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21.1 Document Organization

Note: All chapters linked below for automated document generation.

21.1.1 Front Matter

- Document Information

21.1.2 Chapter 1: Overview

- Architecture
- Top-Level Port List
- Clocks and Reset

21.1.3 Chapter 2: Functional Blocks

Macro/Integration Level: - Stream Core - Scheduler Group Array - Scheduler Group

Control Path: - Scheduler - Descriptor Engine

Read Datapath: - AXI Read Engine - Stream Alloc Control

SRAM Buffering: - SRAM Controller - SRAM Controller Unit - Stream Latency Bridge

Write Datapath: - Stream Drain Control - AXI Write Engine

Configuration & Monitoring: - APB Config - Performance Profiler - MonBus AXIL Group

21.1.4 Chapter 3: Interfaces

- Interface Overview
- AXI4 Interface Specification
- AXIL4 Interface Specification
- APB Programming Interface
- MonBus Interface Specification

21.1.5 Chapter 4: Registers

- Register Map
- Complete Register Reference (generated from the RDL)

21.1.6 Chapter 5: Programming

- Programming Guide

21.1.7 Chapter 6: Configuration Reference

- Complete Configuration Guide

21.2 Quick Reference

21.2.1 Functional Unit Blocks (FUB)

Module	File	Purpose	Lines	Status
descriptor_engine	`rtl/fub/descriptor_engine.sv`	Descriptor fetch/parse (256-bit), one per channel (`SHARED_ENGINES=0`)	~300	[Done]
descriptor_engine_shared	`rtl/fub/descriptor_engine_shared.sv`	One descriptor fetch engine for every channel (`SHARED_ENGINES=1`, default; ported from RAPIDS)	~600	[Done]
scheduler	`rtl/fub/scheduler.sv`	Transfer coordinator	~400	[Done] Created (corrected)
axi_read_engine	`rtl/fub/axi_read_engine.sv`	AXI read master	~350	[Done] Created (no FSM - streaming pipeline)
axi_write_engine	`rtl/fub/axi_write_engine.s`	AXI write master	~400	[Done] Created (no FSM -

Module	File	Purpose	Lines	Status
	v			streaming pipeline)
sram_controller	rtl/fub/sram_controller.sv	Per-channel buffer management	~350	[Done] Created (no FSM - pointer arithmetic + pre-allocation)
sram_controller_unit	rtl/fub/sram_controller_unit.sv	Per-channel buffer unit (wraps gaxi_fifo_sync; replaced the old simple_sram.sv)	~300	[Done]
perf_profiler	rtl/fub/perf_profiler.sv	Channel performance profiling	~400	[Done] Dual-mode timestamp/elapsed tracking

21.2.2 Integration Blocks (MAC)

Module	File	Purpose	Lines	Status
scheduler_group_array	rtl/macro/scheduler_group_array.sv	NUM_CHANNELS (8 or 16) scheduler array + descriptor fetch, shared or per channel (SHARED_ENGINES)	~500	[Done]
apb_config	rtl/macro/stream_regs.rdl + generated wrapper (regs/generated/)	Config registers	~350	[Done] PeakRDL-based

Module	File	Purpose	Lines	Status
monbus_axil_group	rtl/amba/ocelus/monbus_axil4_axil4_group.sv (shared AMBA monitor block)	MonBus + AXIL	~800	[Done]
stream_top	rtl/top/stream_top_ch8.sv	Top-level (8-channel)	~500	[Done]
stream_top_ch16	rtl/top/stream_top_ch16.sv	Top-level (16-channel); generated from stream_top_ch8.sv by bin/gen_stream_regs_16ch.py	~500	[Done]
apb_config (16ch)	rtl/macro/generated_16ch/stream_regs_16ch.rdl (derived) + regs/generated_16ch/	16-channel config registers	-	[Done] PeakRDL-based

21.3 Performance Modes (AXI Engines)

STREAM AXI engines support three performance modes via compile-time parameters, offering area/performance trade-offs from tutorial simplicity to datacenter throughput.

21.3.1 Holistic Overview: V1/V2/V3 Working Together

Design Philosophy: - **V1 (Low):** Tutorial-focused, simple architecture, minimal area - **V2 (Medium):** Command pipelining hides memory latency, 6.7x throughput improvement - **V3 (High):** Out-of-order completion maximizes memory controller efficiency, 7.0x throughput

Key Insight: The benefit scales with memory latency. For low-latency SRAM (2-3 cycles), V1 achieves 40% throughput. For high-latency DDR4 (70-100 cycles), V1 drops to 14% but V2/V3 maintain 94-98% throughput.

Parameterization Strategy: - `ENABLE_CMD_PIPELINE = 0`: V1 (default, tutorial mode) - `ENABLE_CMD_PIPELINE = 1`: V2 (command pipelined, best area efficiency) - `ENABLE_CMD_PIPELINE = 1, ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read): V3 (out-of-order, maximum throughput)

21.3.2 V1 - Low Performance (Single Outstanding Transaction)

Architecture: Single-burst-per-request, blocking on completion - **Area:** ~1,250 LUTs per engine - **Throughput:** 0.14 beats/cycle (DDR4), 0.40 beats/cycle (SRAM) - **Outstanding Txns:** 1 - **Use Case:** Tutorial examples, embedded systems, low-latency SRAM - **Key Feature:** Zero-bubble streaming pipeline (NO FSM!)

Blocking Behavior: - Write: Blocks on B response before accepting next request - Read: Blocks on R last before accepting next request - Simple control: 3 flags (`r_ar_inflight`, `r_ar_valid`, completion tracking)

21.3.3 V2 - Medium Performance (Command Pipelined)

Architecture: Command queue decouples command acceptance from data completion - **Area:** ~2,000 LUTs per engine (1.6x increase) - **Throughput:** 0.94 beats/cycle (DDR4), 0.85 beats/cycle (SRAM) - **Outstanding Txns:** 4-8 (command queue depth) - **Use Case:** General-purpose FPGA, DDR3/DDR4, balanced area/performance - **Key Feature:** Hides memory latency via pipelining (6.7x improvement)

Command Pipelining: - Write: AW queue + W drain FSM + B scoreboard (async response tracking) - Read: AR queue + R in-order reception (simpler than write, no B channel) - Per-command SRAM pointers enable multiple bursts from same channel

Best Area Efficiency: 4.19x throughput per unit area (6.7x throughput / 1.6x area)

21.3.4 V3 - High Performance (Out-of-Order Completion)

Architecture: OOO command selection maximizes memory controller efficiency - **Area:** ~3,500-4,000 LUTs per engine (2.8-3.2x increase) - **Throughput:** 0.98 beats/cycle (DDR4), 0.92 beats/cycle (SRAM) - **Outstanding Txns:** 8-16 (larger command queue) - **Use Case:** Datacenter, ASIC, HBM2, high-performance memory controllers - **Key Feature:** Flexible drain order based on SRAM data availability (7.0x improvement)

Out-of-Order Mechanisms: - Write: OOO W drain (select command with SRAM data ready), AXI ID matching for B responses - Read: OOO R reception (match m_axi_rid to queue entry), independent SRAM write per command - Transaction ID structure: `{counter, channel_id}` preserves channel routing for MonBus

Why OOO is Naturally Supported: 1. Per-channel SRAM partitioning (no cross-channel hazards) 2. Per-command pointer tracking (no pointer collision) 3. Transaction ID matching (channel ID in lower bits for routing)

21.3.5 Performance Comparison Summary

Configuration	Area	DDR4 Throughput	SRAM Throughput	Area Efficiency	Use Case
V1	1.0x	0.14 beats/cycle (1.0x)	0.40 beats/cycle (1.0x)	1.00	Tutorial, embedded
V2	1.6x	0.94 beats/cycle (6.7x)	0.85 beats/cycle (2.1x)	4.19	General FPGA
V3	2.8x	0.98 beats/cycle (7.0x)	0.92 beats/cycle (2.3x)	2.50	Datacenter, ASIC

Key Takeaway: V2 offers best area efficiency (4.19x) for typical FPGA use cases. V3 provides marginal throughput improvement (7.0x vs 6.7x) at higher area cost, justified only for high-performance memory controllers that support out-of-order responses.

21.4 Clock and Reset Summary

21.4.1 Clock Domains

Clock	Frequency	Usage
`aclk`	100-500 MHz	Primary - all STREAM logic, AXI/AXIL interfaces
`pclk`	50-200 MHz	APB configuration interface (may be async to aclk)

21.4.2 Reset Signals

Reset	Polarity	Type	Usage
`aresetn`	Active-low	Async assert, sync deassert	Primary - all STREAM logic
`presetn`	Active-low	Async assert, sync deassert	APB configuration interface

See: Clocks and Reset for complete timing specifications

21.5 Interface Summary

21.5.1 External Interfaces

Interface	Type	Width	Purpose
APB	Slave	32-bit	Configuration registers
AXI (Descriptor)	Master	256-bit	Descriptor fetch
AXI (Read)	Master	512-bit (param)	Source data read
AXI (Write)	Master	512-bit (param)	Destination data write
AXIL (Slave)	Slave	32-bit	Error/interrupt FIFO access
AXIL (Master)	Master	32-bit	MonBus packet logging to memory
IRQ	Output	1-bit	Error interrupt

21.5.2 Internal Buses

Interface	Width	Purpose
MonBus	64-bit	Internal monitoring bus (channels -> monbus_axil_group)

21.6 Area Estimates

21.6.1 By Performance Mode

Configuration	Total LUTs	SRAM	Use Case
Low (Tutorial)	~9,500	64 KB	Educational, area-constrained
Medium (Typical)	~11,200	64 KB	Balanced FPGA implementations
High (Performance)	~13,700	64 KB	High-throughput ASIC/FPGA

21.6.2 Breakdown (Low Performance)

Component	Instances	Area/Instance	Total
Descriptor Engine	8	~300 LUTs	~2,400 LUTs
Scheduler	8	~400 LUTs	~3,200 LUTs
AXI Read Engine (Low)	1	~250 LUTs	~250 LUTs
AXI Write Engine (Low)	1	~250 LUTs	~250 LUTs
SRAM Controller	1	~1,600 LUTs	~1,600 LUTs
Simple SRAM (internal)	1-8	1024x64B total	64 KB
Channel Arbiter	3	~150 LUTs	~450 LUTs
APB Config	1	~350 LUTs	~350 LUTs
MonBus AXIL Group	1	~1,000 LUTs	~1,000 LUTs
Total	-	-	**~9,500 LUTs** +

Component	Instances	Area/Instance	Total
			64KB

21.7 Related Documentation

- **PRD.md** - Product requirements and overview
- **CLAUDE.md** - AI development guide
- **Register Generation** - PeakRDL workflow

21.8 Specification Conventions

21.8.1 Signal Naming

- **Clock:** `aclk`, `pclk`
- **Reset:** `aresetn`, `presetn` (active-low)
- **Valid/Ready:** Standard AXI/custom handshake
- **Registers:** `r_` prefix (e.g., `r_state`, `r_counter`)
- **Wires:** `w_` prefix (e.g., `w_next_state`, `w_grant`)

21.8.2 Parameter Naming

- **Uppercase:** `NUM_CHANNELS`, `DATA_WIDTH`, `ADDR_WIDTH`
- **String parameters:** `PERFORMANCE` ("LOW", "MEDIUM", "HIGH")

21.8.3 State Machine Naming

```
typedef enum logic [3:0] {
    IDLE   = 4'h0,
    ACTIVE = 4'h1,
    // ...
} state_t;

state_t r_state, w_next_state;  // Current and next state
```

Last Updated: 2026-05-11 **Maintained By:** STREAM Architecture Team

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22 Product Requirements Document (PRD)

22.1 STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory

Version: 1.0 **Date:** 2025-10-17 **Status:** Nearly Complete - Final Integration Pending **Owner:** RTL Design Sherpa Project **Parent Document:** `/PRD.md`

22.2 1. Executive Summary

The **STREAM** (Scatter-gather Transfer Rapid Engine for AXI Memory) is a simplified DMA-style accelerator designed for efficient memory-to-memory data movement with descriptor-based scatter-gather support. STREAM serves as a beginner-friendly tutorial demonstrating descriptor-based DMA engine design patterns while maintaining production-quality RTL practices.

22.2.1 1.1 Quick Stats

- **Modules:** ~8-10 SystemVerilog files (estimated)
- **Channels:** Maximum 8 independent channels
- **Interfaces:** APB (config), AXI4 (descriptor fetch + data read/write), MonBus (monitoring)
- **Architecture:** Simplified from RAPIDS - pure memory-to-memory
- **Tutorial Focus:** Aligned addresses only, straightforward data flow
- **Status:** Nearly complete - config interface and top-level wrapper pending

22.2.2 1.2 Project Goals

- **Primary:** Educational DMA engine demonstrating scatter-gather descriptor chains
- **Secondary:** Production-quality RTL suitable for FPGA/ASIC implementation
- **Tertiary:** Foundation for understanding more complex DMA architectures (e.g., RAPIDS)

22.3 2. Key Design Principles

22.3.1 2.1 Simplifications from RAPIDS

STREAM is intentionally simplified for tutorial purposes:

Feature	RAPIDS	STREAM
Network Interfaces	Yes (Network master/slave)	✗ No (pure memory-to-memory)
Address Alignment	Complex fixup logic	√ **Aligned addresses only**
Credit Management	Exponential encoding	√ Simple transaction limits
Control Engines	Control read/write engines	✗ No (direct APB config)
Descriptor Length	Chunks (4-byte)	√ **Beats** (data width)
Program Engine	Complex alignment FSM	√ Simplified coordination

22.3.2 2.2 Tutorial-Friendly Features

- **Aligned Addresses:** Source and destination addresses must be aligned to data width
- **Length in Beats:** Descriptor length specified in data beats (not bytes/chunks)
- **Single APB Write:** One APB register write kicks off entire descriptor chain
- **No Circular Buffers:** Chained descriptors with explicit termination
- **Parameterized Engines:** Multiple AXI engine versions (compile-time selection)

22.4 3. Architecture Overview

22.4.1 3.1 Top-Level Block Diagram

```
STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory)
├── APB Config Slave
│   └── Channel registers (8 channels, kick-off via write)
│
├── Descriptor Fetch
│   └── AXI Master (Read) - Fetch descriptors from memory
│
├── Scheduler Group (shared across all channels)
│   ├── Descriptor Engine   (FIFO, parsing - FROM RAPIDS)
│   ├── Scheduler           (Simplified FSM coordination)
│   └── Channel Arbitration (8 independent channels)
│
```

```
├── Data Path (shared resources)
│   ├── AXI Master (Read)  - Parameterized data width
│   ├── Simple SRAM        - Dual-port buffer (FROM RAPIDS)
│   └── AXI Master (Write) - Parameterized data width
│
└── MonBus Reporter
    └── MonBus Master - 64-bit monitoring packets
```

22.4.2 3.2 Data Flow

Descriptor-Based Transfer Sequence:

```
1. Software writes to APB channel register
   ↓
2. Descriptor fetch via AXI descriptor master
   ↓
3. Descriptor Engine parses descriptor fields
   ↓
4. Scheduler coordinates data transfer:
   a. AXI Read Engine fetches source data → SRAM buffer
   b. AXI Write Engine writes SRAM → destination
   ↓
5. Check for chained descriptor (next_descriptor_ptr != 0)
   ↓ (if chained)
6. Fetch next descriptor, repeat from step 3
   ↓ (if last)
7. Generate MonBus completion packet
```

Channel Independence: - 8 channels operate independently - All channels share: SRAM, AXI data masters, descriptor fetch master - Arbitration required for shared resources

---

22.5 4. Interfaces

22.5.1 4.1 External Interfaces

Interface	Type	Width	Purpose	Notes
APB Slave	Slave	32-bit	Configuration, channel kick-off	Write to channel register starts transfer
AXI Master (Descriptor)	Master	256-bit	Fetch descriptors from memory	Dedicated descriptor fetch path

Interface	Type	Width	Purpose	Notes
AXI Master (Data Read)	Master	Parameterizable	Read source data	Multiple engine versions (compile-time)
AXI Master (Data Write)	Master	Parameterizable	Write destination data	Multiple engine versions (compile-time)
MonBus Master	Master	64-bit	Monitor packet output	Standard AMBA format

22.5.2 4.2 Descriptor Format

256-bit Descriptor Structure:

Bits	Field	Description
[63:0]	`src_addr`	Source address (64-bit, must be aligned to data width)
[127:64]	`dst_addr`	Destination address (64-bit, must be aligned to data width)
[159:128]	`length`	Transfer length in **BEATS** (not bytes!)
[191:160]	`next_descriptor_ptr`	Address of next descriptor (0 = last in chain)
[192]	`valid`	Descriptor is valid
[193]	`interrupt`	Generate interrupt on completion
[194]	`last`	Last descriptor in chain (explicit flag)
[195]	`error`	Error status (used for reporting)
[199:196]	`channel_id`	Channel ID (0-7)
[207:200]	`priority`	Transfer priority (for arbitration)
[255:208]	`reserved`	Reserved for future use

Key Descriptor Features: - ✓ **Chained descriptors:** `next_descriptor_ptr` links to next descriptor - **✗ No circular buffers:** Explicit termination (last flag or ptr=0) - ✓ **Length in beats:**

Simplified for tutorial (no byte/chunk conversion) - ✓ **Aligned addresses:** Tutorial constraint (performance hidden for now)

22.6 5. Key Components

22.6.1 5.1 Descriptor Engine (APB-Only for STREAM)

Source: Adapted from RAPIDS `descriptor_engine.sv`

Purpose: - Autonomous descriptor fetch and chaining - APB interface for initial descriptor address - AXI read interface for descriptor memory fetches - Descriptor FIFO storage and distribution

Key Features: - ✓ **Autonomous chaining:** Automatically fetches next descriptor if `next_descriptor_ptr != 0` AND `last == 0` - ✓ **Address validation:** Validates next descriptor addresses against `cfg_addr0/1_base/limit` - ✓ **APB blocking:** APB blocked until `channel_idle == 1` (channel fully idle) - ✓ **Error handling:** AXI errors stop chaining, set `descriptor_error`, block `descriptor_valid`

Adaptations from RAPIDS: - ✓ **APB-only:** Single APB write kicks off entire descriptor chain - ✓ **Descriptor Read Address FIFO:** 2-deep FIFO stores addresses for AXI fetch (APB + chaining) - ✓ **Chaining logic:** Descriptor engine autonomously manages `next_descriptor_ptr` chaining

Idle Signal: - `descriptor_engine_idle` asserted when: - FSM in `RD_IDLE` state - No pending descriptor fetches (address FIFO empty) - No active AXI transactions

22.6.2 5.2 Scheduler Group (Integration Wrapper)

Purpose: Wraps descriptor engine and scheduler into a single channel processing unit

Architecture:

```
scheduler_group (
    // APB interface (from APB config slave)
    .apb_valid         (apb_valid),
    .apb_ready         (apb_ready),   // Blocked when channel not idle
    .apb_addr          (descriptor_addr),

    // Channel idle signal composition (CRITICAL!)
    .channel_idle      (channel_idle),

    // Descriptor → Scheduler flow
    .descriptor_valid  (desc_valid),
    .descriptor_ready  (desc_ready),
    .descriptor_packet (desc_packet),
```

```
    // Data engine interfaces
    .datard_*           (datard_*),    // Read engine
    .datawr_*           (datawr_*),    // Write engine

    // Status
    .scheduler_idle     (sched_idle),
    .descriptor_idle    (desc_idle)
);
```

Channel Idle Signal Composition:

```
// Channel is idle ONLY when BOTH sub-blocks are idle
assign channel_idle = scheduler_idle && descriptor_engine_idle;
```

Why Both Signals Matter:

Signal	Indicates	Used For
`scheduler_idle`	No active data transfers, all descriptors processed	Prevents new APB request during active transfer
`descriptor_engine_idle`	No pending descriptor fetches (FIFO empty)	Prevents new APB request during chaining
`channel_idle` (AND of both)	Channel fully quiescent	**Gates APB interface**

APB Blocking Logic:

```
// Descriptor engine blocks APB when channel not idle
assign apb_ready = apb_skid_ready_in &&
                   !r_channel_reset_active &&
                   w_desc_addr_fifo_empty &&    // No pending fetches
                   channel_idle;                 // Scheduler +
descriptor idle
```

Example Scenario:

```
1. Software writes APB → descriptor_addr = 0x1000
   - channel_idle = 1 (both idle)
   - APB accepted

2. Descriptor engine fetches descriptor @ 0x1000
   - descriptor_engine_idle = 0 (fetch in progress)
   - channel_idle = 0
   - APB BLOCKED

3. Descriptor pushed to scheduler
   - descriptor_engine_idle = 1 (fetch complete)
   - scheduler_idle = 0 (transfer starting)
```

```
   - channel_idle = 0
   - APB BLOCKED

4. Scheduler completes data transfer
   - Descriptor has next_descriptor_ptr = 0x1100 (chained!)
   - Descriptor engine autonomously fetches @ 0x1100
   - descriptor_engine_idle = 0 (autonomous fetch)
   - channel_idle = 0
   - APB BLOCKED

5. Final descriptor completes (last = 1 OR next_ptr = 0)
   - scheduler_idle = 1 (transfer done)
   - descriptor_engine_idle = 1 (no more fetches)
   - channel_idle = 1
   - APB UNBLOCKED (ready for next transfer!)
```

Key Insight: The AND gate ensures software cannot interrupt a descriptor chain in progress!

22.6.3 5.3 Scheduler (Simplified from RAPIDS)

Purpose: - Coordinate descriptor-to-data-transfer flow - Manage 8 independent channels - Arbitrate shared resources (SRAM, AXI masters)

FSM States:

```
typedef enum logic [7:0] {
    SCHED_IDLE              = 8'b00000001,  // Idle, waiting for
channel activation
    SCHED_FETCH_DESCRIPTOR  = 8'b00000010,  // Fetch descriptor via
AXI master
    SCHED_PARSE_DESCRIPTOR  = 8'b00000100,  // Parse descriptor fields
    SCHED_READ_PHASE        = 8'b00001000,  // Coordinate read engine
    SCHED_WRITE_PHASE       = 8'b00010000,  // Coordinate write engine
    SCHED_CHAIN_CHECK       = 8'b00100000,  // Check for next
descriptor
    SCHED_COMPLETE          = 8'b01000000,  // Transfer complete,
report status
    SCHED_ERROR             = 8'b10000000   // Error state
} scheduler_state_t;
```

Key Differences from RAPIDS: - ✗ No credit management (just simple transaction limits) - ✗ No program engine coordination (no alignment fixup) - √ Simplified FSM (no control read/write phases)

22.6.4 5.3 AXI Read Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming reads from memory to SRAM buffer

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: FSMs are horrible for performance! Instead, use: - **Arbiter** selects which channel's `datard_*` interface gets access - **Streaming pipeline** continuously moves data when granted - **Data interface** (`datard_valid`, `datard_ready`, `datard_beats_remaining`) controls flow

Data Read Interface (per channel):

```
// Channel requests read access
input  logic        datard_valid;          // Channel has read request
output logic        datard_ready;          // Engine ready for request
input  logic [63:0] datard_addr;           // Source address (aligned)
input  logic [31:0] datard_beats_remaining; // Beats left to read
input  logic [7:0]  datard_burst_len;      // Preferred burst length
input  logic [3:0]  datard_channel_id;     // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding read, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding reads, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

```
1. Arbiter grants channel access based on priority
2. Engine accepts datard_* request (continuous streaming)
3. AXI AR channel issues read burst
4. AXI R channel streams data → SRAM (no FSM stalls!)
5. Engine updates beats_remaining, accepts next request
6. Different channels can have different burst lengths (e.g., CH0: 8
beats, CH1: 16 beats)
```

Example: Different Burst Lengths per Channel

```
// Channel 0 prefers 8-beat bursts
datard_burst_len[0] = 8'd8;

// Channel 1 prefers 16-beat bursts
datard_burst_len[1] = 8'd16;

// Engine adapts to requested burst length (within MAX_BURST_LEN)
```

22.6.5 5.4 AXI Write Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming writes from SRAM buffer to memory

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: Same as read engine - no FSMs! Use streaming pipeline with arbiter.

Data Write Interface (per channel):

```
// Channel requests write access
input  logic        datawr_valid;          // Channel has write
```

```
request
output logic        datawr_ready;           // Engine ready for
request
input  logic [63:0] datawr_addr;            // Destination address
(aligned)
input  logic [31:0] datawr_beats_remaining; // Beats left to write
input  logic [7:0]  datawr_burst_len;       // Preferred burst length
input  logic [3:0]  datawr_channel_id;      // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding write, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding writes, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

```
1. Arbiter grants channel access based on priority
2. Engine accepts datawr_* request (continuous streaming)
3. Engine reads data from SRAM
4. AXI AW channel issues write address
5. AXI W channel streams data (no FSM stalls!)
6. AXI B channel receives response, updates beats_remaining
7. Different channels can have different burst lengths (e.g., CH0: 8
beats, CH2: 32 beats)
```

Read/Write Asymmetry Example:

```
// Channel can use different burst lengths for read vs write
// Example: Read in small bursts, write in large bursts
datard_burst_len[0] = 8'd8;   // Read: 8 beats
datawr_burst_len[0] = 8'd16;  // Write: 16 beats

// Engine handles asymmetry via SRAM buffering
```

22.6.6 5.5 SRAM Buffering

Source: Originally a direct copy of RAPIDS `simple_sram.sv`; now implemented by `rtl/fub/sram_controller.sv` + `sram_controller_unit.sv` (wrapping the shared `gaxi_fifo_sync` primitive)

Purpose: - Dual-port SRAM buffer - Decouples read and write engines - Shared across all channels (arbitration required)

Why Reuse: - Standard dual-port SRAM design - Proven in RAPIDS integration tests - Parameterizable depth and width

---

22.7 6. Configuration and Control

22.7.1 6.1 APB Register Map

Offset	Register	Access	Description
0x0000	GLOBAL_CTRL	RW	Global enable, reset
0x0004	GLOBAL_STATUS	RO	Global status, error flags
0x0100	CH0_CTRL	WO	**Channel 0 kick-off** (write descriptor address)
0x0104	CH0_STATUS	RO	Channel 0 status
0x0108	CH0_DESC_ADDR	RO	Channel 0 current descriptor address
0x010C	CH0_BYTES_XFER	RO	Channel 0 bytes transferred
...	...	...	... (repeat for channels 1-7)
0x0200	CH1_CTRL	WO	Channel 1 kick-off
...			
0x0700	CH7_CTRL	WO	Channel 7 kick-off

Kick-Off Sequence: 1. Software writes descriptor address to CHx_CTRL register 2. STREAM fetches descriptor from memory 3. Transfer begins automatically 4. If chained, STREAM follows next_descriptor_ptr automatically 5. Completion reported via MonBus packet

22.7.2 6.2 Channel Configuration

Each channel independently configurable: - **Descriptor start address:** Written to CHx_CTRL - **Priority:** Encoded in descriptor (arbitration) - **Interrupt enable:** Per-descriptor flag - **Status monitoring:** Read CHx_STATUS

22.8 7. Resource Sharing and Arbitration

22.8.1 7.1 Shared Resources

All channels share: 1. **Descriptor Fetch AXI Master** - Fetches descriptors for all channels 2. **Data Read AXI Master** - Reads source data for all channels 3. **Data Write AXI Master** - Writes

destination data for all channels 4. **SRAM Buffer** - Shared buffer (dual-port, but still arbitrated) 5. **MonBus Reporter** - Single monitor output

22.8.2 7.2 Arbitration Strategy

Priority-Based Round-Robin: - Channels have priority field in descriptor - Higher priority = serviced first - Within same priority: round-robin - Prevents starvation with timeout

Example Arbitration:

```
Channel 0: Priority 7 (highest)
Channel 1: Priority 5
Channel 2: Priority 5
Channel 3: Priority 3
```

Service order: CH0 → CH1 → CH2 (round-robin) → CH0 → CH1 → CH2 → CH3 ...

22.9 8. Error Detection and Recovery

22.9.1 8.1 Error Types

Error Type	Detection	Response
Invalid descriptor	Valid bit = 0	Skip, move to next
Alignment error	Address not aligned	Set error flag, halt channel
AXI SLVERR	AXI response	Set error flag, halt channel
AXI DECERR	AXI response	Set error flag, halt channel
Timeout	Transaction timeout	Set error flag, halt channel
SRAM overflow	Buffer full	Backpressure, wait

22.9.2 8.2 Error Recovery

Per-Channel Error Handling: - Error sets channel to `CH_ERROR` state - Channel halts, does not affect other channels - Software must: 1. Read `CHx_STATUS` to identify error 2. Clear error condition 3. Re-kick channel with new descriptor

No Automatic Retry: - Tutorial design keeps error handling simple - Software responsible for retry logic

22.10 9. MonBus Integration

22.10.1 9.1 Standard MonBus Format

Uses standard 64-bit MonBus packet format: - [63:60] Packet Type (0=ERROR, 1=COMPL, etc.) - [59:57] Protocol (custom STREAM protocol) - [56:53] Event Code (STREAM-specific events) - [52:47] Channel ID (0-7) - [46:43] Unit ID (unused for STREAM) - [42:35] Agent ID (unused for STREAM) - [34:0] Event Data (address, byte count, etc.)

22.10.2 9.2 STREAM Event Codes

Code	Event	Description
0x0	`DESC_START`	Descriptor started
0x1	`DESC_COMPLETE`	Descriptor completed
0x2	`READ_START`	Read phase started
0x3	`READ_COMPLETE`	Read phase completed
0x4	`WRITE_START`	Write phase started
0x5	`WRITE_COMPLETE`	Write phase completed
0x6	`CHAIN_FETCH`	Chained descriptor fetch
0xF	`ERROR`	Error occurred

22.10.3 9.3 Default Configuration

Tutorial-Friendly Defaults: - **Errors only:** `cfg_error_enable = 1`, all others = 0 - **Interrupts:** Descriptor flag controls per-transfer interrupt - **Reduces MonBus traffic** for beginner understanding

22.11 10. Design Constraints

22.11.1 10.1 Tutorial Constraints (Intentional Simplifications)

Constraint	Rationale
Aligned addresses only	Simplify data path, hide alignment

Constraint	Rationale
	complexity
Length in beats	Avoid byte/chunk conversion math
No circular buffers	Explicit termination easier to understand
Single APB kick-off	Simple software model
No credit management	Avoid exponential encoding complexity

22.11.2 10.2 Implementation Constraints

Parameter	Value	Notes
Max channels	8	Compile-time parameter
Max burst length	256	AXI4 spec limit
Descriptor size	256 bits	4 × 64-bit words
SRAM depth	Parameterizable	Typical: 1024-4096 entries
Data width	Parameterizable	Typical: 512-bit

22.12 11. Verification Strategy

22.12.1 11.1 Test Organization

```
projects/components/dma-ip/stream/dv/tests/
├── fub/                        # Functional Unit Block tests
│   ├── test_descriptor_engine.py
│   ├── test_scheduler.py
│   ├── test_sram_controller.py (+ test_sram_controller_alloc.py)
│   ├── test_perf_profiler.py
│   └── test_stream_latency_bridge.py
├── macro/                      # Multi-block scenarios (stream_core,
datapaths)
└── top/                        # Top-level (stream_top_ch8) tests
```

22.12.2 11.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

---

22.13 12. Performance Characteristics

22.13.1 12.1 Throughput by Engine Version

V1 (Low Performance - Tutorial Mode): - **Throughput:** 0.14 beats/cycle (DDR4), 0.40 beats/cycle (SRAM) - **Architecture:** Single outstanding transaction, blocks on completion - **Use Case:** Tutorial examples, embedded systems, low-latency SRAM

V2 (Medium Performance - Command Pipelined): - **Throughput:** 0.94 beats/cycle (DDR4), 0.85 beats/cycle (SRAM) - **Improvement:** 6.7x over V1 (DDR4), 2.1x over V1 (SRAM) - **Architecture:** Command queue (4-8 deep), hides memory latency - **Use Case:** General-purpose FPGA, DDR3/DDR4, best area efficiency

V3 (High Performance - Out-of-Order): - **Throughput:** 0.98 beats/cycle (DDR4), 0.92 beats/cycle (SRAM) - **Improvement:** 7.0x over V1 (DDR4), 2.3x over V1 (SRAM) - **Architecture:** OOO command selection, maximizes memory controller efficiency - **Use Case:** Datacenter, ASIC, HBM2, high-performance memory

Key Insight: Benefit scales with memory latency. V1 throughput degrades from 40% (SRAM) to 14% (DDR4), while V2/V3 maintain 94-98% throughput regardless of latency.

Configuration Parameters: - `ENABLE_CMD_PIPELINE = 0`: V1 (default, tutorial mode) - `ENABLE_CMD_PIPELINE = 1`: V2 (command pipelined) - `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read): V3

Factors Affecting Throughput: - Memory latency (V2/V3 hide latency via pipelining) - SRAM buffer size (limits burst pipelining) - Channel arbitration overhead - Descriptor fetch latency - Engine version (V1/V2/V3 configuration)

22.13.2 12.2 Latency

Transfer Latency Breakdown: - Descriptor fetch: ~10-50 cycles (depends on memory) - Read phase: `(length / burst_len) × burst_latency` - Write phase: `(length / burst_len) × burst_latency` - End-to-end: Typically <200 cycles for small transfers (V1), <100 cycles (V2/V3 pipelined)

V2/V3 Latency Hiding: - Command pipelining overlaps descriptor fetch, read, write operations - Multiple outstanding transactions hide memory latency - OOO completion (V3) reduces head-of-line blocking

22.13.3 12.3 Resource Utilization by Engine Version

V1 Configuration (Tutorial - Minimum Area): - Total: ~9,500 LUTs + 64 KB SRAM - Descriptor Engine (8×): ~2,400 LUTs - Scheduler (8×): ~3,200 LUTs - AXI Read Engine: ~1,250 LUTs - AXI Write Engine: ~1,250 LUTs - SRAM Controller: ~1,600 LUTs - APB Config: ~350 LUTs - MonBus AXIL Group: ~1,000 LUTs

V2 Configuration (Balanced - Best Area Efficiency): - Total: ~11,000 LUTs + 64 KB SRAM (1.16x area) - AXI Read Engine: ~2,000 LUTs (1.6x increase, 6.7x throughput) - AXI Write Engine: ~2,500 LUTs (2.0x increase, 6.7x throughput) - Other blocks: Same as V1

V3 Configuration (Maximum Performance): - Total: ~14,000 LUTs + 64 KB SRAM (1.47x area) - AXI Read Engine: ~3,500 LUTs (2.8x increase, 7.0x throughput) - AXI Write Engine: ~4,000 LUTs (3.2x increase, 7.0x throughput) - Other blocks: Same as V1

Area Efficiency Comparison: - V1: 1.00 throughput / 1.00 area = 1.00 - V2: 6.70 throughput / 1.16 area = 5.78 (best efficiency) - V3: 7.00 throughput / 1.47 area = 4.76

Recommendation: V2 provides best area efficiency for most use cases. V3 justified only for high-performance memory controllers that support OOO responses.

22.14 13. Development Roadmap

22.14.1 13.1 Phase 1: Foundation (Current)

- ✓ Directory structure
- ✓ Package definitions (`stream_pkg.sv`)
- ✓ Imports header (`stream_imports.svh`)
- ⏳ Documentation (this PRD)

22.14.2 13.2 Phase 2: Core Blocks

- Adapt descriptor engine from RAPIDS
- Design simplified scheduler
- Create APB config interface
- Copy simple SRAM from RAPIDS

22.14.3 13.3 Phase 3: Data Path

- AXI read engine (version 1 - basic)
- AXI write engine (version 1 - basic)
- SRAM integration
- Channel arbitration

22.14.4 13.4 Phase 4: Integration

- Top-level module
- MonBus reporter
- Integration testbench
- Single-channel validation

22.14.5 13.5 Phase 5: Multi-Channel

- 8-channel support
- Arbiter implementation
- Multi-channel tests
- Performance tuning

22.14.6 13.6 Phase 6: Advanced Engines (Future - V2/V3)

Goal: Add parameterized high-performance engine variants

V2 - Command Pipelined (Medium Performance): - Command queue implementation (4-8 deep) - W drain FSM for write engine - B response scoreboard (write) or in-order R reception (read) - Per-command SRAM pointer tracking - Parameter: `ENABLE_CMD_PIPELINE = 1` - Expected: 6.7x throughput improvement over V1

V3 - Out-of-Order Completion (High Performance): - OOO command selection logic - Transaction ID matching (AXI ID to queue entry) - SRAM data availability checking (write engine) - R beat matching to queue entry (read engine) - Parameters: `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read) - Expected: 7.0x throughput improvement over V1

Deliverables: - Updated RTL with parameterization - Performance comparison tests (V1 vs V2 vs V3) - Tutorial documentation explaining trade-offs - Area/throughput measurements on target FPGA

22.15 14. Educational Value

22.15.1 14.1 Learning Objectives

STREAM teaches: 1. **Descriptor-based DMA design patterns** - Descriptor structure and parsing - Chained descriptors (scatter-gather) - Descriptor fetch via AXI

2. **AXI4 Memory Interface Usage**
   - Burst transactions
   - Address/data/response channels

    - Outstanding transactions
3. **Resource Sharing and Arbitration**
    - Multiple channels sharing AXI masters
    - Priority-based arbitration
    - Conflict resolution
4. **FSM Design and Coordination**
    - Multiple interconnected FSMs
    - State machine composition
    - Error handling
5. **Buffer Management**
    - SRAM-based buffering
    - Rate matching
    - Flow control

22.15.2 14.2 Progression Path

Learning Sequence: 1. **STREAM (this project):** Memory-to-memory DMA, aligned addresses 2. **STREAM Extended:** Add alignment fixup, more complex scenarios 3. **RAPIDS:** Add network interfaces, credit management, full complexity

---

22.16 15. Success Criteria

22.16.1 15.1 Functional

- ✓ Single descriptor transfer working
- ✓ Chained descriptors (2-3 deep)
- ✓ Multi-channel operation (8 channels concurrent)
- ✓ Error detection and reporting
- ✓ MonBus packet generation
- ✓ >90% functional test coverage

22.16.2 15.2 Quality

- ✓ Verilator compiles with 0 warnings
- ✓ All tests passing (100% success rate)
- ✓ Comprehensive documentation
- ✓ Tutorial-quality code comments

22.16.3 15.3 Performance

- ✓ Achieves >80% of theoretical AXI bandwidth
- ✓ <5K LUT utilization (excluding SRAM)
- ✓ <200 cycle end-to-end latency for small transfers

22.17 16. Open Questions (For Review)

22.17.1 16.1 Descriptor Engine Adaptation

- **Q:** Should descriptor engine use APB-only, or support additional modes?
- **A (pending):** TBD - depends on software use case preference

22.17.2 16.2 AXI Descriptor Master

- **Q:** Fixed 256-bit width, or parameterizable?
- **A (pending):** Propose fixed 256-bit for simplicity

22.17.3 16.3 Channel Arbitration

- **Q:** Fixed priority, or dynamic priority based on age/fairness?
- **A (pending):** Propose fixed priority with round-robin for tutorial simplicity

22.17.4 16.4 SRAM Partitioning

- **Q:** Single shared SRAM, or per-channel SRAMs?
- **A (pending):** Propose single shared SRAM with arbitration (matches RAPIDS pattern)

22.18 16.2 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

```
/mnt/data/github/rtldesignsherpa/projects/components/dma-ip/stream/
docs/
```

Quick Command: Use the provided shell scripts:

```
cd projects/components/dma-ip/stream/docs
./generate_has_pdf.sh   # STREAM HAS
./generate_mas_pdf.sh   # STREAM MAS
```

The shell scripts will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_has/ and stream_mas/ index files 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

22.19 17. References

22.19.1 17.1 Internal Documentation

- **RAPIDS PRD:** `projects/components/dma-ip/rapids/PRD.md` - Parent architecture
- **RAPIDS Descriptor Engine:** `projects/components/dma-ip/rapids/rtl/fub_beats/descriptor_engine_beats.sv` (beats rearchitecture of the original)
- **Shared SRAM/FIFO primitives:** `rtl/amba/gaxi/gaxi_fifo_sync.sv`, `rtl/amba/shared/sdpram_core.sv` (the old `simple_sram.sv` was removed)
- **AMBA PRD:** `docs/markdown/rtl-amba/index.md` - MonBus integration
- **Repository Guide:** `/CLAUDE.md` - Design patterns and conventions

22.19.2 17.2 External References

- **AXI4 Specification:** ARM IHI0022E
- **APB Specification:** ARM IHI0024C
- CocoTB Documentation: https://docs.cocotb.org/
- Verilator Manual: https://verilator.org/guide/latest/

Document Version: 1.0 **Last Updated:** 2025-10-17 **Review Cycle:** Weekly during initial design **Next Review:** TBD (after user feedback) **Owner:** RTL Design Sherpa Project

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22.20 Navigation

- **← Back to Root:** `/PRD.md`
- **Parent Architecture:** `projects/components/dma-ip/rapids/PRD.md`
- **AI Guidance:** `CLAUDE.md` (to be created)
- **Quick Start:** `README.md` (to be created)

23 Claude Code Guide: STREAM Subsystem

Version: 1.0 **Last Updated:** 2025-10-17 **Purpose:** AI-specific guidance for working with STREAM subsystem

---

23.1 Quick Context

What: STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory **Status:** ✓ Complete - tutorial-focused DMA engine (fub/macro/top all passing; on-board characterization flows exist) **Your Role:** Help users build a beginner-friendly descriptor-based DMA engine

📖 **Complete Specification:** `projects/components/dma-ip/stream/PRD.md` ← **Always reference this for technical details**

23.2 📖 Global Requirements Reference

IMPORTANT: Review `/GLOBAL_REQUIREMENTS.md` for all mandatory requirements

All mandatory requirements are consolidated in the global requirements document: - **See:** `/GLOBAL_REQUIREMENTS.md` - Repository-wide mandatory requirements - **STREAM-Specific:** Attribution format, tutorial focus (intentional simplifications) - **Universal:** TB location, three methods, TBBase inheritance, 100% success

This CLAUDE.md provides STREAM-specific guidance. Also review: - Root `/CLAUDE.md` - Repository-wide patterns - `projects/components/CLAUDE.md` - Project area standards (reset macros, FPGA attributes) - `bin/TBClasses/` - Shared TB framework (full CocoTBFramework in the RTLDesignSherpa-DV repo)

23.3 Critical Rules for This Subsystem

23.3.1 Rule #0.1: TUTORIAL FOCUS - Intentional Simplifications

⚠️ STREAM is INTENTIONALLY SIMPLIFIED for educational purposes ⚠️

Key Simplifications (DO NOT "fix" these): 1. ✓ **Aligned addresses only** - No alignment fixup logic (kept for RAPIDS) 2. ✓ **Length in beats** - Not bytes or chunks (simplifies math) 3. ✓ **No circular buffers** - Explicit chain termination only 4. ✓ **No credit management** - Simple transaction limits 5. ✓ **Pure memory-to-memory** - No network interfaces

When users ask "Can we add alignment fixup?": - ✓ **Correct answer:** "STREAM intentionally keeps addresses aligned for tutorial simplicity. For complex alignment, see RAPIDS." - ✗ **Wrong answer:** "Sure, let me add alignment logic..." (defeats tutorial purpose!)

23.3.2 Rule #0.1: Testbench Location and Test Structure (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.1 for complete requirement

STREAM-Specific Directory Structure:

```
projects/components/dma-ip/stream/dv/
├── tbclasses/                    # ⭐ STREAM TB classes (project
area!)
│   ├── scheduler_tb.py           # Scheduler testbench
│   ├── descriptor_engine_tb.py   # Descriptor engine testbench
│   ├── sram_controller_tb.py     # SRAM controller testbench
│   ├── stream_core_tb.py         # Macro-level core testbench
│   └── stream_top_tb.py          # Top-level testbench
└── tests/                        # Test runners import from
tbclasses/
    ├── fub/                      # Per-block tests
    ├── macro/                    # Integration tests
    └── top/                      # Top-level tests
```

STREAM Import Pattern:

```python
# Import STREAM TB from project area
from projects.components.dma_ip.stream.dv.tbclasses.scheduler_tb
import SchedulerTB

# Shared utilities from framework
from TBClasses.shared.tbbase import TBBase
```

📖 **Complete Pattern:** `projects/components/dma-ip/rapids/CLAUDE.md` Rule #0.1

23.3.3 Rule #0.2: Three Mandatory TB Methods (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.2 for complete requirement

STREAM-Specific Context:

STREAM has simpler config requirements than RAPIDS - most config can be set after reset.

Standard STREAM Pattern:

```python
class SchedulerTB(TBBase):
    async def setup_clocks_and_reset(self):
        """Standard STREAM initialization"""
        await self.start_clock('clk', freq=10, units='ns')
        await self.assert_reset()
```

```
        await self.wait_clocks('clk', 10)
        await self.deassert_reset()
        await self.wait_clocks('clk', 5)

    async def assert_reset(self):
        self.dut.rst_n.value = 0

    async def deassert_reset(self):
        self.dut.rst_n.value = 1
```

Note: Unlike RAPIDS, STREAM typically doesn't need config set before reset.

23.3.4 Rule #1: REUSE from RAPIDS Where Appropriate

Direct Reuse (No Modification): - ✓ `descriptor_engine.sv` - Works with STREAM descriptors - ✓ Shared FIFO/SRAM primitives (`gaxi_fifo_sync`, used inside `sram_controller_unit.sv`; the old `simple_sram.sv` was removed) - ✓ Descriptor engine tests - Adapt imports only

Adapt from RAPIDS: - ⚠️ `scheduler.sv` - **Simplify FSM** (no credit management, no control engines) - ⚠️ AXI engines - **Create simplified versions** (no alignment fixup)

Create New for STREAM: - 📝 APB config interface - PeakRDL-generated (like HPET), 8 channels, kick-off registers - 📝 Top-level integration - Different interface set

Always Ask Yourself: "Can I reuse from RAPIDS instead of creating new?"

23.3.5 Rule #2: Descriptor Format is DIFFERENT from RAPIDS

STREAM Descriptor (256-bit): - `src_addr` (64-bit), `dst_addr` (64-bit), `length` (**beats**), `next_descriptor_ptr` (32-bit) - **Length is in BEATS, not chunks!**

RAPIDS Descriptor: - Uses chunks (4-byte units) - Has alignment metadata

When comparing/referencing RAPIDS: - ✓ "RAPIDS uses chunks, STREAM uses beats for tutorial simplicity" - ✗ Don't assume RAPIDS descriptor format applies to STREAM

23.3.6 Rule #3: Know the Shared Resources

All channels (8 or 16, by top) share: 1. Descriptor fetch AXI master – and, by default (`SHARED_ENGINES=1`), ONE `descriptor_engine_shared` that fetches for every channel (ARID = channel). `SHARED_ENGINES=0` restores one `descriptor_engine` per channel behind an AR arbiter. 2. Data read AXI master 3. Data write AXI master 4. SRAM buffer 5. MonBus reporter

Arbitration is required! - Never assume exclusive access - All shared resources need arbiter logic - Priority-based round-robin (descriptor priority field)

23.4 Architecture Quick Reference

23.4.1 Block Organization

```
STREAM Architecture (as implemented)
├── rtl/fub/                          # Functional unit blocks
│   ├── descriptor_engine.sv          # Descriptor fetch/parse, one per
channel (SHARED_ENGINES=0)
│   ├── descriptor_engine_shared.sv # One fetch engine for all
channels (default; ported from RAPIDS)
│   ├── scheduler.sv                  # Simplified per-channel scheduler
│   ├── axi_read_engine.sv            # Data read engine
│   ├── axi_write_engine.sv           # Data write engine
│   ├── sram_controller.sv            # SRAM buffering (+
sram_controller_unit.sv per channel)
│   ├── stream_alloc_ctrl.sv          # Space allocation tracking
│   ├── stream_drain_ctrl.sv          # Drain control
│   ├── stream_latency_bridge.sv      # Latency decoupling
│   ├── (moved) stream_run_addr_gen.sv -> projects/components/utility-
ip/misc/rtl/  # shared with RAPIDS
│   └── perf_profiler.sv              # Performance profiling
│
├── rtl/macro/                        # Integration blocks
│   ├── scheduler_group.sv            # Per-channel scheduler group
│   ├── scheduler_group_array.sv      # NUM_CHANNELS array + descriptor
fetch (SHARED_ENGINES)
│   ├── stream_core.sv                # Core integration
│   ├── stream_regs.rdl               # PeakRDL register source (8-
channel, hand-written)
│   └── generated_16ch/               # 16-channel RDL, DERIVED by
bin/gen_stream_regs_16ch.py
│
└── rtl/top/                          # Top level
    ├── stream_top_ch8.sv             # 8-channel top-level wrapper
(hand-written)
    ├── stream_top_ch16.sv            # 16-channel top, GENERATED from
stream_top_ch8.sv
    ├── generated/regsel_{8,16}ch/  # regblock/hwif/kick glue
includes; the filelist picks one
    └── stream_config_block.sv        # Configuration block
```

16 channels: edit the 8-channel sources (`stream_regs.rdl`, `stream_mon_regs.rdl`, `stream_top_ch8.sv`), then run `python3 bin/gen_stream_regs_16ch.py` and regenerate BOTH regblocks with `bin/peakrdl_generate.py`; `--check` and `bin/check_rdl_regen.py` catch a stale 16-channel copy. Software and tests go by register name (`rtl/stream_regmap.py` / `rtl/stream_regmap_16ch.py`): the 16-channel map moves CH_STATE to 0x180 and OBS_CTRL.CAT_SEL to [5:4].

Monitoring uses the shared monbus group modules from `rtl/amba/ocelus/` (e.g. `monbus_axil4_axil4_group.sv`) rather than a STREAM-local reporter.

23.5 Common User Questions and Responses

23.5.1 Q: "How is STREAM different from RAPIDS?"

A: STREAM is intentionally simplified for tutorial purposes:

Simplifications: | Feature | RAPIDS | STREAM | |———|———|———| | **Interfaces** | APB + AXI + Network | APB + AXI only | | **Address Alignment** | Complex fixup | Aligned only | | **Credit Management** | Exponential encoding | Simple limits | | **Descriptor Length** | Chunks (4-byte) | Beats (data width) | | **Control Engines** | Control read/write | None (direct APB) |

Learning Path: 1. **STREAM** - Basic DMA with scatter-gather 2. **STREAM Extended** - Add alignment fixup 3. **RAPIDS** - Full complexity with network + credit management

📖 **See:** `PRD.md` Section 2.1 for complete comparison table

23.5.2 Q: "How do I kick off a transfer?"

A: Single APB write starts descriptor chain:

```
// Software writes descriptor address to channel register
write_apb(ADDR_CH0_CTRL, 0x1000_0000);  // Start address of descriptor

// STREAM automatically:
// 1. Fetches descriptor from 0x1000_0000
// 2. Parses src_addr, dst_addr, length
// 3. Reads source data → SRAM
// 4. Writes SRAM → destination
// 5. Checks next_descriptor_ptr
//    - If != 0: Fetch next descriptor, repeat
//    - If == 0 or last flag set: Complete
```

Chained Descriptors:

```
Descriptor 0 @ 0x1000_0000:
  src_addr = 0x2000_0000
  dst_addr = 0x3000_0000
  length = 64 beats
  next_descriptor_ptr = 0x1000_0100  ← Points to next descriptor

Descriptor 1 @ 0x1000_0100:
  src_addr = 0x2000_1000
  dst_addr = 0x3000_1000
```

```
  length = 32 beats
  next_descriptor_ptr = 0x0000_0000  ← Last descriptor (0 = stop)
```

📖 **See:** PRD.md Section 3.2 for complete data flow

23.5.3 Q: "How many channels can I use?"

A: 8 or 16 independent channels, by top (`stream_top_ch8` / `stream_top_ch16`):

Channel Operation: - Each channel has own FSM state - Each channel can have separate descriptor chain - All channels share: AXI masters, SRAM, MonBus

Arbitration: - Priority-based (descriptor priority field) - Round-robin within same priority - Prevents starvation with timeout

Example:

```
// Kick off 3 channels concurrently
write_apb(ADDR_CH0_CTRL, desc0_addr);  // Channel 0: Priority 7
write_apb(ADDR_CH1_CTRL, desc1_addr);  // Channel 1: Priority 5
write_apb(ADDR_CH2_CTRL, desc2_addr);  // Channel 2: Priority 5

// Service order: CH0 → CH1 → CH2 → CH0 → CH1 → CH2 ...
```

📖 **See:** PRD.md Section 7 for arbitration details

23.5.4 Q: "What's the descriptor format?"

A: 256-bit descriptor (4 × 64-bit words):

```
typedef struct packed {
    logic [63:0]  reserved;             // [255:192] Reserved
    logic [7:0]   priority;             // [207:200] Priority
    logic [3:0]   channel_id;           // [199:196] Channel ID
    logic         error;                // [195] Error flag
    logic         last;                 // [194] Last in chain
    logic         interrupt;            // [193] Generate interrupt
    logic         valid;                // [192] Valid descriptor
    logic [31:0]  next_descriptor_ptr;  // [191:160] Next descriptor
address
    logic [31:0]  length;               // [159:128] Length in BEATS
★
    logic [63:0]  dst_addr;             // [127:64] Destination
address
    logic [63:0]  src_addr;             // [63:0] Source address
} descriptor_t;
```

★ **CRITICAL:** `length` is in **BEATS**, not bytes or chunks!

Example:

```
Data width = 512 bits = 64 bytes
Length = 16 beats
Total transfer = 16 × 64 = 1024 bytes
```

📖 **See:** PRD.md Section 4.2 for complete descriptor specification

23.5.5 Q: "How do I run STREAM tests?"

A: Multi-layered test approach (same as RAPIDS):

```
# 1. FUB (Functional Unit Block) Tests - Individual blocks
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_descriptor_engine.
py -v
pytest projects/components/dma-ip/stream/dv/tests/fub/ -v  # All FUB
tests

# 2. Macro / Top Tests - Multi-block scenarios
pytest projects/components/dma-ip/stream/dv/tests/macro/ -v
pytest projects/components/dma-ip/stream/dv/tests/top/ -v

# Run with waveforms for debugging
WAVES=1 pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
```

Test Organization: - **FUB tests:** Focus on individual block functionality (`dv/tests/fub/`) - **Macro/top tests:** Verify block-to-block interfaces and complete data flows (`dv/tests/macro/`, `dv/tests/top/`)

23.6 Integration Patterns

23.6.1 Pattern 1: Basic STREAM Instantiation

```
stream_top #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .ADDR_WIDTH(64),
    .SRAM_DEPTH(4096)
) u_stream (
    // Clock and Reset
    .aclk               (system_clk),
    .aresetn            (system_rst_n),

    // APB Configuration Interface
```

```
    .s_apb_paddr        (apb_paddr),
    .s_apb_psel         (apb_psel),
    .s_apb_penable      (apb_penable),
    .s_apb_pwrite       (apb_pwrite),
    .s_apb_pwdata       (apb_pwdata),
    .s_apb_pready       (apb_pready),
    .s_apb_prdata       (apb_prdata),
    .s_apb_pslverr      (apb_pslverr),

    // AXI Master - Descriptor Fetch (256-bit)
    .m_axi_desc_araddr  (desc_araddr),
    .m_axi_desc_arlen   (desc_arlen),
    .m_axi_desc_arsize  (desc_arsize),
    .m_axi_desc_arvalid (desc_arvalid),
    .m_axi_desc_arready (desc_arready),
    .m_axi_desc_rdata   (desc_rdata),
    .m_axi_desc_rresp   (desc_rresp),
    .m_axi_desc_rlast   (desc_rlast),
    .m_axi_desc_rvalid  (desc_rvalid),
    .m_axi_desc_rready  (desc_rready),

    // AXI Master - Data Read (parameterizable width)
    .m_axi_rd_araddr    (rd_araddr),
    // ... (full AXI4 AR + R channels)

    // AXI Master - Data Write (parameterizable width)
    .m_axi_wr_awaddr    (wr_awaddr),
    // ... (full AXI4 AW + W + B channels)

    // MonBus Output
    .monbus_pkt_valid   (stream_mon_valid),
    .monbus_pkt_ready   (stream_mon_ready),
    .monbus_pkt_data    (stream_mon_data)
);
```

23.6.2 Pattern 2: Descriptor Creation (Software Model)

```
// C/C++ software model for descriptor creation
typedef struct {
    uint64_t src_addr;          // Source address (must be aligned!)
    uint64_t dst_addr;          // Destination address (must be
aligned!)
    uint32_t length;            // Transfer length in BEATS
    uint32_t next_descriptor_ptr; // Next descriptor address (0 =
last)
    uint8_t  control;           // valid | interrupt | last | error |
channel_id | priority
} stream_descriptor_t;
```

```
// Create descriptor chain
stream_descriptor_t desc[2];

// Descriptor 0
desc[0].src_addr = 0x80000000ULL;  // Aligned to 64B (512-bit data)
desc[0].dst_addr = 0x90000000ULL;
desc[0].length = 64;  // 64 beats × 64 bytes = 4KB transfer
desc[0].next_descriptor_ptr = (uint32_t)&desc[1];  // Chain to next
desc[0].control = 0x01;  // valid = 1

// Descriptor 1 (last)
desc[1].src_addr = 0x80001000ULL;
desc[1].dst_addr = 0x90001000ULL;
desc[1].length = 32;  // 32 beats × 64 bytes = 2KB transfer
desc[1].next_descriptor_ptr = 0;  // Last descriptor
desc[1].control = 0x45;  // valid | last | interrupt

// Kick off transfer
write_apb_reg(CH0_CTRL, (uint32_t)&desc[0]);
```

23.6.3 Pattern 3: MonBus Integration

```
// Always add downstream FIFO for MonBus
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_stream_mon_fifo (
    .axi_aclk   (aclk),
    .axi_aresetn(aresetn),
    // producer side: wr_ready is "not full", so it is what
backpressures MonBus
    .wr_valid   (monbus_pkt_valid),
    .wr_data    (monbus_pkt_data),
    .wr_ready   (monbus_pkt_ready),
    // consumer side
    .rd_valid   (fifo_mon_valid),
    .rd_data    (fifo_mon_data),
    .rd_ready   (consumer_ready)
);
```

23.7 Anti-Patterns to Catch

23.7.1 ✗ Anti-Pattern 1: Adding Alignment Fixup

✗ WRONG:
"Let me add alignment fixup logic to handle unaligned addresses..."

✓ CORRECTED:
"STREAM intentionally requires aligned addresses for tutorial simplicity.
If you need unaligned transfers, that's a great learning exercise for extending STREAM, or use RAPIDS which has full alignment support."

23.7.2 ✗ Anti-Pattern 2: Using Length in Bytes

✗ WRONG:
```
descriptor.length = 4096;  // Thinking this is 4096 bytes
```

✓ CORRECTED:
```
// Length is in BEATS, not bytes!
// For 512-bit data width (64 bytes per beat):
descriptor.length = 4096 / 64;  // = 64 beats for 4096 bytes
```

23.7.3 ✗ Anti-Pattern 3: Circular Buffer Descriptors

✗ WRONG:
```
// Descriptor chain that loops back
desc[9].next_descriptor_ptr = &desc[0];  // Circular!
```

✓ CORRECTED:
"STREAM does not support circular buffers (no stop condition).
Always terminate chains explicitly:
```
  desc[last].next_descriptor_ptr = 0;  // Stop
  desc[last].last = 1;  // Explicit last flag
```

23.7.4 ✗ Anti-Pattern 4: Assuming Exclusive Channel Access

✗ WRONG:
```
// Assume channel has exclusive AXI master access
assign m_axi_arvalid = channel_request;  // No arbitration!
```

✓ CORRECTED:
```
// All channels share AXI masters - arbitration required
channel_arbiter u_arbiter (
    .ch_requests (channel_requests[7:0]),
    .ch_grant    (channel_grant[7:0]),
    .axi_master_available (axi_master_idle)
);
```

23.8 Debugging Workflow

23.8.1 Issue: Descriptor Not Fetched

Check in order: 1. ✓ APB write to CHx_CTRL register successful? 2. ✓ Descriptor address valid? 3. ✓ AXI descriptor master not stalled? 4. ✓ Descriptor memory accessible? 5. ✓ Arbiter granting descriptor fetch?

Debug commands:

```
pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v -s
# Verbose test
WAVES=1 pytest
projects/components/dma-ip/stream/dv/tests/fub/test_scheduler.py -v
gtkwave <logs dir>/*.fst  # Inspect FSM state transitions
```

23.8.2 Issue: Data Transfer Stalls

Check in order: 1. ✓ SRAM buffer depth sufficient? 2. ✓ Source/destination addresses aligned? 3. ✓ AXI read/write engines getting grants? 4. ✓ Downstream AXI ready signals asserted? 5. ✓ Channel not in error state?

Waveform Analysis: - Check SRAM read/write pointers - Verify AXI AR/AW/R/W/B handshakes - Inspect scheduler FSM state - Check arbiter grant signals

23.8.3 Issue: Chained Descriptors Not Following

Check in order: 1. ✓ `next_descriptor_ptr` != 0? 2. ✓ `last` flag not set prematurely? 3. ✓ Descriptor fetch completing successfully? 4. ✓ Scheduler transitioning to CHAIN_CHECK state? 5. ✓ MonBus showing CHAIN_FETCH event?

23.9 Testing Guidance

23.9.1 Test Organization

```
projects/components/dma-ip/stream/dv/tests/
├── fub/                          # Individual block tests
│   ├── test_descriptor_engine.py
│   ├── test_scheduler.py
│   ├── test_sram_controller.py (+ test_sram_controller_alloc.py)
│   ├── test_perf_profiler.py
│   └── test_stream_latency_bridge.py
├── macro/                        # Multi-block scenarios (stream_core,
```

```
scheduler_group, datapaths)
└── top/                          # Top-level (stream_top_ch8) tests
```

23.9.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

---

23.10 Key Documentation Links

23.10.1 Always Reference These

This Subsystem: - `projects/components/dma-ip/stream/PRD.md` - **Complete specification** - `projects/components/dma-ip/stream/README.md` - Quick start guide - `projects/components/dma-ip/stream/known_issues/` - Bug tracking - `projects/components/dma-ip/stream/docs/stream_has/`, `docs/stream_mas/` - HAS/MAS spec trees

Related: - `projects/components/dma-ip/rapids/PRD.md` - Parent architecture (for comparison) - `docs/markdown/rtl-amba/index.md` - MonBus integration - `/PRD.md` - Master requirements - `/CLAUDE.md` - Repository guide

---

23.11 Quick Commands

```bash
# View complete specification
cat projects/components/dma-ip/stream/PRD.md

# Check package definition
cat projects/components/dma-ip/stream/rtl/includes/stream_pkg.sv

# Run tests
pytest projects/components/dma-ip/stream/dv/tests/fub/ -v
pytest projects/components/dma-ip/stream/dv/tests/macro/ -v
pytest projects/components/dma-ip/stream/dv/tests/top/ -v

# Lint
verilator --lint-only
```

```
projects/components/dma-ip/stream/rtl/fub/scheduler.sv

# Search for modules
find projects/components/dma-ip/stream/rtl/ -name "*.sv" -exec grep -H
"^module" {} \;
```

23.12 Remember

1. 📖 **Tutorial focus** - Intentional simplifications for learning
2. 🔄 **Reuse from RAPIDS** - Descriptor engine, SRAM, patterns
3. 📏 **Length in beats** - Not bytes or chunks!
4. 🔗 **Aligned addresses** - No fixup logic
5. ⛓️ **Chained descriptors** - No circular buffers
6. 🎯 **8 or 16 channels** - Shared resources, arbitration required
7. 🔍 **MonBus standard** - Same format as AMBA/RAPIDS
8. 🏗️ **Testbench reuse** - Always create TB classes in `projects/components/dma-ip/stream/dv/tbclasses/` (project area, not the framework)

23.13 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

`/mnt/data/github/rtldesignsherpa/projects/components/dma-ip/stream/docs/`

Quick Command: Use the provided shell scripts:

```
cd projects/components/dma-ip/stream/docs
./generate_has_pdf.sh   # STREAM HAS
./generate_mas_pdf.sh   # STREAM MAS
```

The shell scripts will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_has/ and stream_mas/ index files 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

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